Regression Discontinuity Design

The Case for RDD

1 min

Research suggests that individuals put more into retirement accounts or pension plans if their employer matches a portion of their contributions.

Imagine that a country's legislature passes a law that requires employers with more than 300 employees to provide a retirement contribution matching program. Using tax data, lawmakers compile a dataset with the following information from each of 200 different companies:

- `size` : Number of employees
- `group` : Contribution Matching Program Group ("No Program" vs. "Program")
- `contribution` : Average monthly employee contributions (in dollars)

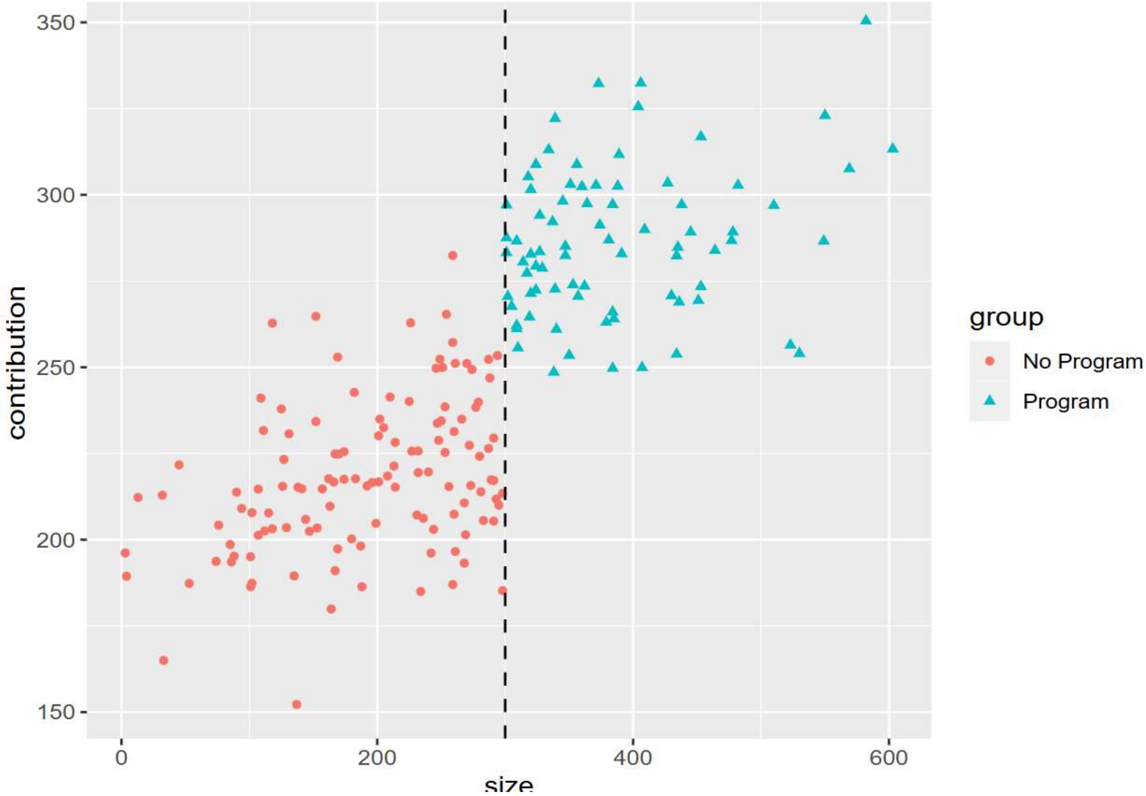
Number of employees — acting as the forcing variable with a cutoff at 300 employees — dictates whether or not a company has a contribution matching program.

We want to assess whether the policy caused an increase in average monthly retirement contributions by employees. This example is a perfect candidate for a new technique, called regression discontinuity design.

RDD works by only focusing on the points near the cutoff. Companies with close to 300 employees are probably very similar to one another. So the ONLY difference, on average, between companies with 299 employees and those with 301 employees should be whether they have the contribution matching program. This means we should have a treatment and control group that look a lot like those of a randomized experiment!

Take a look at the plot in the learning environment, which shows the contribution matching program data. The number of employees (forcing variable) is on the x-axis, and the average monthly contribution (outcome variable) is on the y-axis.

Do you think that companies with 5 employees are similar to companies with 600? What about companies with 200 employees versus 400 employees?



Forcing Variables in RDD

1 min

We need a little more vocabulary before we can dive into more details about Regression Discontinuity Design.

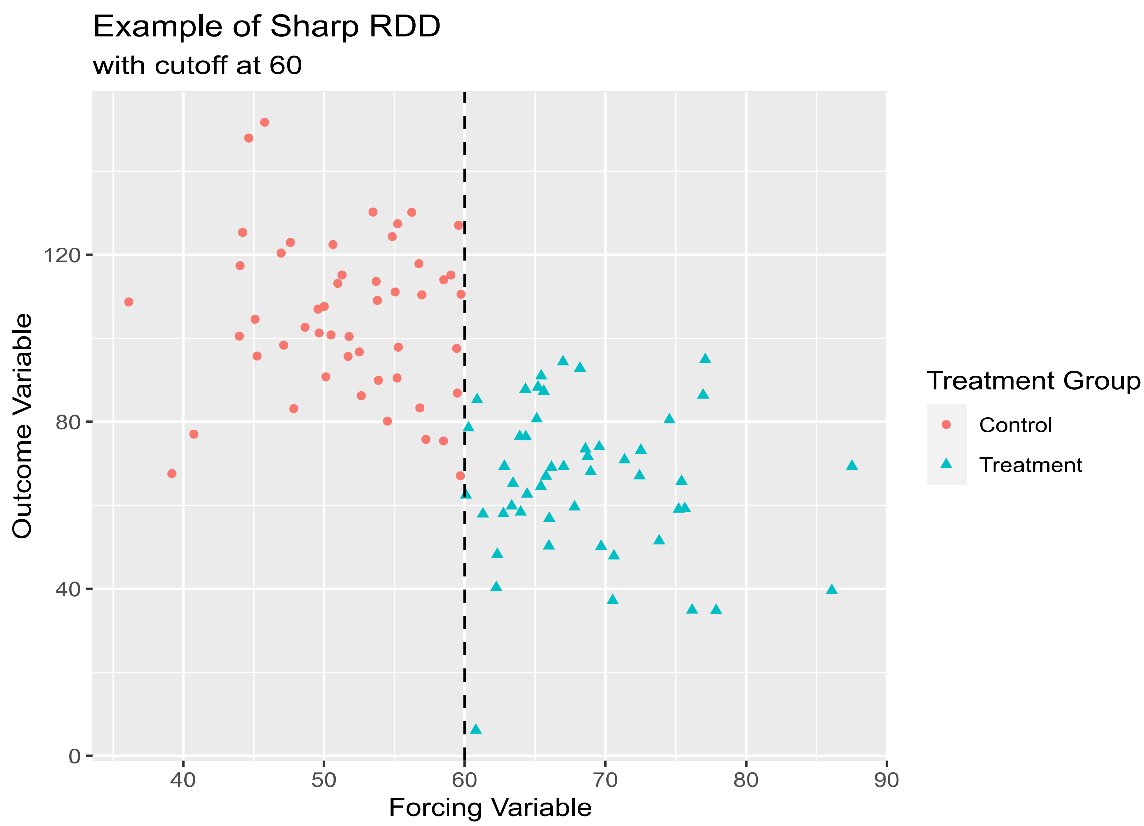
Sometimes treatment group assignment is dictated by one continuous variable known as a *forcing variable* (the company size in our example). The forcing variable, also referred to as a *rating variable* or *running variable*, has a cutoff value such that:

- Individuals with a value smaller than the cutoff are in one treatment group.
- Individuals with a value larger than the cutoff are in the other treatment group.

The treatment group is perfectly predicted by the forcing variable. In this scenario, we cannot rely on other causal inference techniques such as matching or weighting methods because there is not a consistent mixture of treatment and controls across different values of the forcing variable.

Instructions

1. Take a look at the plot provided in the learning environment. Note how points with a value below 60 on the x-axis belong to the control group, while those with values above 60 belong to the treatment group.



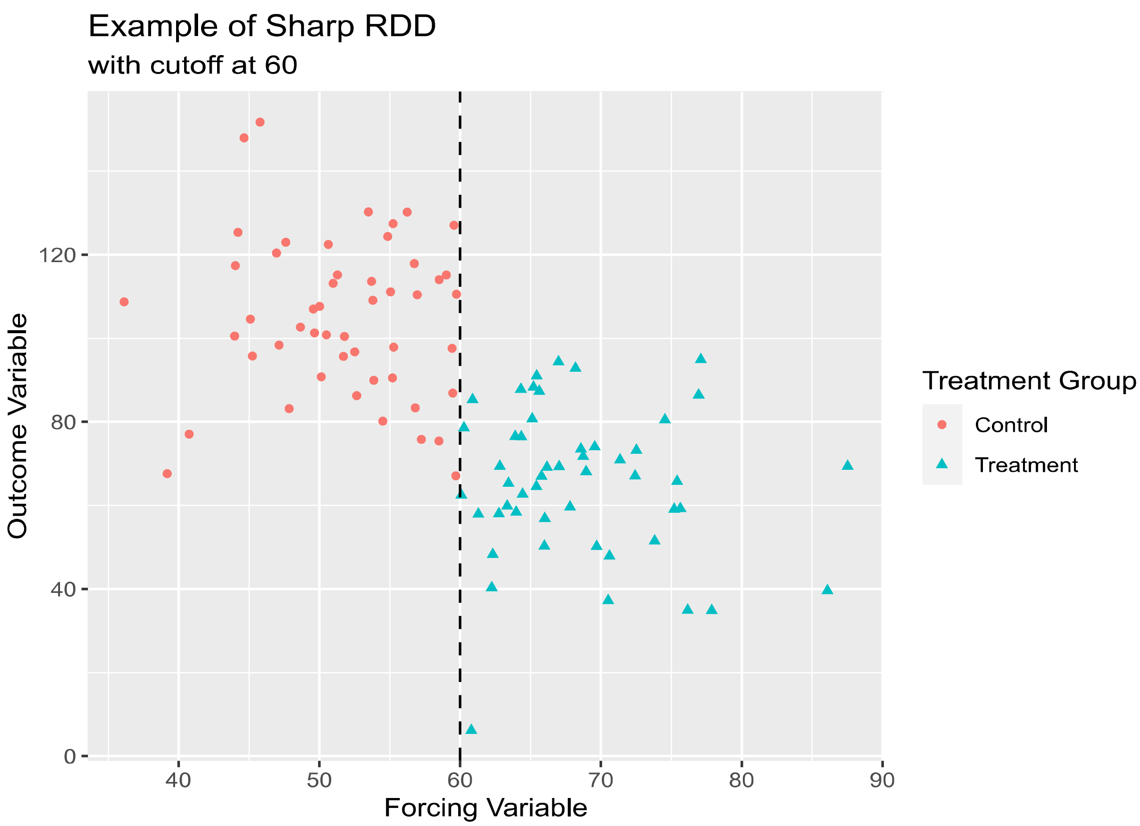
Regression Discontinuity Design

Sharp or Fuzzy?

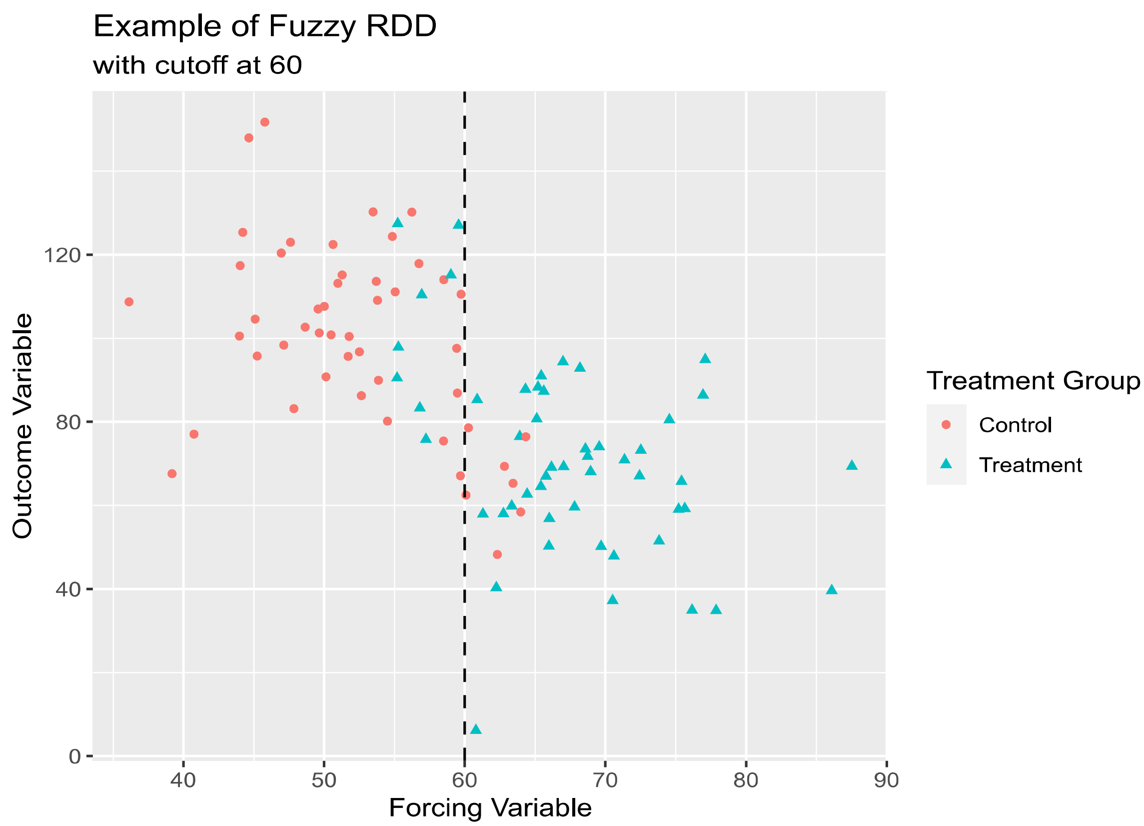
2 min

The forcing variable cutpoint can either be exact or not exact:

- If the cutpoint is exact, the probability of treatment changes from zero to one at the cutpoint. In other words, all observations on one side of the cutpoint are in the treatment group (and actually received the treatment) and all observations on the other side of the cutpoint are in the control group (and didn't receive the treatment). This is known as a *sharp design*.



- If the cutpoint is NOT exact, the probability of treatment doesn't jump from zero to one at the cutpoint. In other words, there are individuals in each treatment group on BOTH sides of the cutpoint. This is known as a *fuzzy design*.

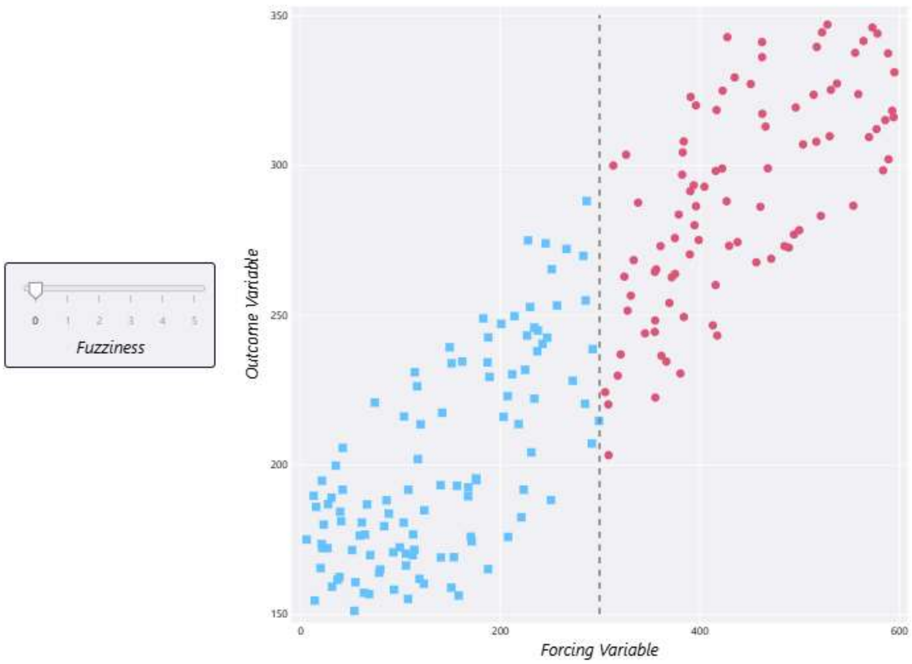


While this distinction may seem minor, it is actually incredibly important to recognize when to use sharp RDD as opposed to fuzzy RDD. Not only do the two approaches make different assumptions about the data, but they also require slightly different statistical methods.

We will focus on sharp RDD in this lesson, even though perfect compliance with the treatment assignment is not always realistic.

Instructions

1. Play around with the interactive plot in the learning environment to see what different levels of “fuzziness” look like in a regression discontinuity design.
2. What does this mean with respect to our companies? As you change the degree of fuzziness, which small companies offer contribution matching? Which large companies are in violation of the law?



Defining RDD Assumptions

2 min

We saw that our employee contribution example requires a sharp regression discontinuity design: all companies with at least 300 employees have a contribution matching program, and all companies with fewer than 300 employees do NOT have a contribution matching program. Before we decide which companies are similar enough to compare, we must consider some assumptions.

In order to get valid estimates of the treatment effect using a sharp RDD approach, several assumptions have to be met.

1. The treatment variable impacts the outcome, but not any of the other [variables](#)
   Preview: Docs Loading link description
   .
2. The treatment assignment happens only at ONE cutpoint value of the forcing variable.
3. Treatment assignment is independent of the potential outcomes within a narrow interval around the cutpoint.
4. Counterfactual outcomes can be modeled within the interval around the cutpoint.

Visual Check

11 min

A scatter plot of our data allows us to check certain RDD conditions visually. We can see whether we have a sharp or fuzzy cutoff. We can also use the plot to check for a *discontinuity* — a sudden change in the outcome variable — at the cutoff.

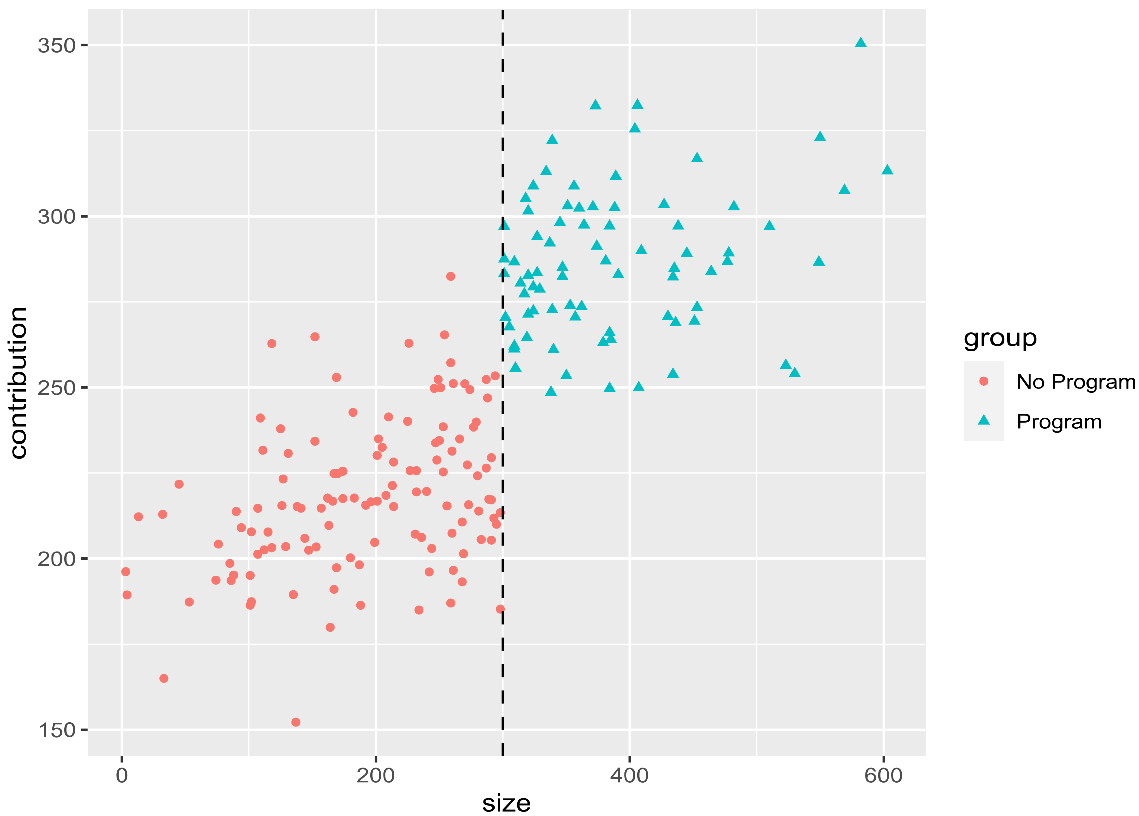
To create this scatter plot with the contribution matching dataset, we can use the `ggplot2` package in R:

```
library(ggplot2) #load ggplot2 package


Copy to Clipboard
```

Our scatter plot should have the number of employees on the x-axis and the contribution amount on the y-axis. The points for treatment and control groups should be different colors and shapes, so we can easily tell the two groups apart. Finally, we add code for a dashed vertical line at the cutoff of 300 employees.

```
# create a scatterplot with treatment groups
ggplot(
  data = cont_data,
  aes(
    x = size, # forcing variable
    y = contribution, # outcome variable
    color = group, # sets point color by treatment group
    shape = group # sets point shape by treatment group
  )) +
  geom_point() +
  geom_vline(xintercept = 300, linetype = "dashed") #add line at 300


Copy to Clipboard
```

This plot clearly shows that we have a sharp RDD, not a fuzzy one. The dashed line at 300 employees separates the two groups into companies that offer a matching program (at least 300) and companies that do NOT offer a matching program (fewer than 300).



We can also check to make sure that there is actually a discontinuity in average contributions based on whether or not companies have more than 300 employees. To do this, we can add a separate best fit line for each treatment group using the `geom_smooth()` function. If we had saved our first plot as `rdd_scatter`, we can add the code for the lines as follows:

```
# add best fit lines for each group to scatter plot
rdd_scatter +
  geom_smooth(
    aes(group = group), #plot separate lines for each group
    method = "lm" #use linear regression
)


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```
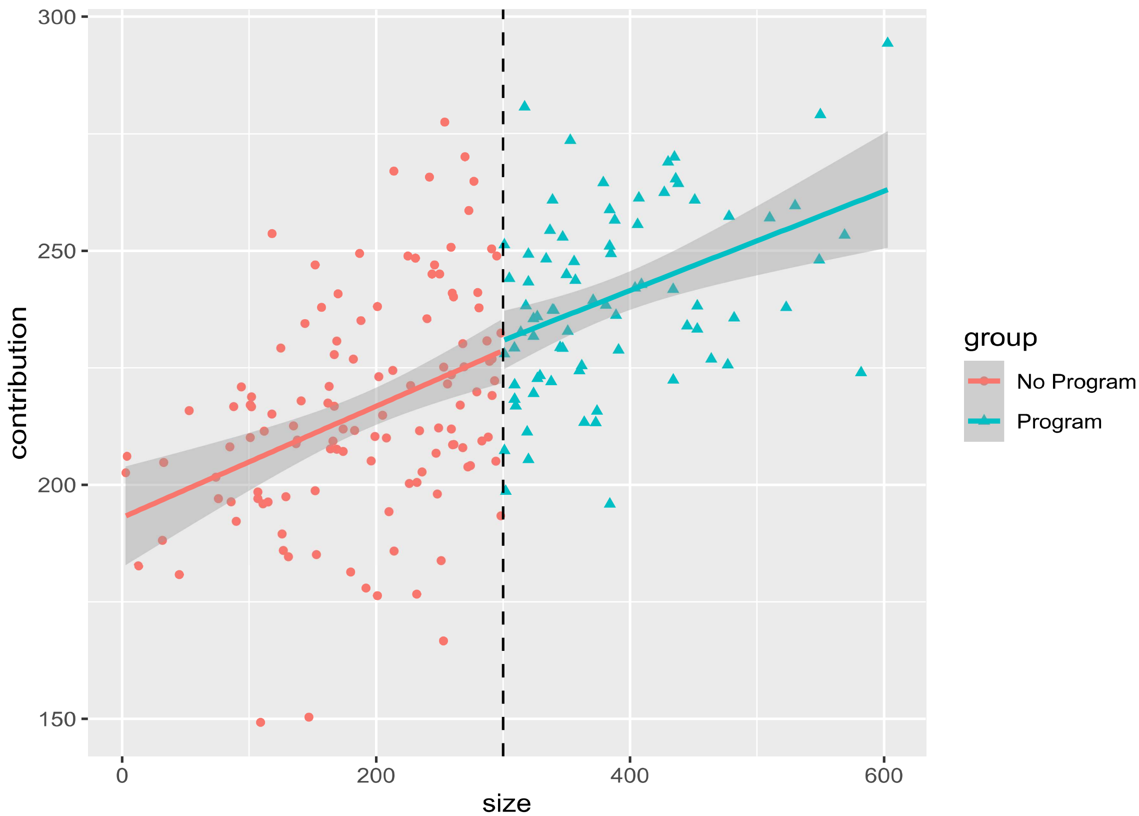

There is an obvious jump in the average contributions at the cutoff point, which means there is a discontinuity present. If there were no discontinuity present, we might see something like this:



Note that there is no jump in the outcome variable here. The lines connect smoothly.

Instructions

1. Checkpoint 1 Enabled
   1.
   A new policy requires that power plants that have an output of 600 megawatts or more have to install an emissions control device that removes harmful chemicals before releasing exhaust into the air. An environmental group decides to measure ambient air quality one mile away from each power plant to assess whether the emissions control device leads to better air quality.
   The `air_data` dataset has been loaded for you in **notebook.Rmd** with the following variables:
   Create a scatter plot named `air_scatter`:
   - `id` : power plant ID.
   - `watts` : power plant output (megawatts)
   - `group` : treatment group formed by cutpoint at 600 watts (control = “No Device”, treatment = “Emissions Device”)
   - `aqi` : air quality index from 0 (good air quality) to 500 (poor air quality)
   - Use `watts` as the forcing variable and `aqi` as the outcome variable.
   - Use different colors and shapes for each treatment group.
   - Add a dashed vertical line at 600 megawatts.

1. A new policy requires that power plants that have an output of 600 megawatts or more have to install an emissions control device that removes harmful chemicals before releasing exhaust into the air. An environmental group decides to measure ambient air quality one mile away from each power plant to assess whether the emissions control device leads to better air quality.
   The `air_data` dataset has been loaded for you in **notebook.Rmd** with the following variables:
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   - `aqi` : air quality index from 0 (good air quality) to 500 (poor air quality)
   - Use `watts` as the forcing variable and `aqi` as the outcome variable.
   - Use different colors and shapes for each treatment group.
   - Add a dashed vertical line at 600 megawatts.
2. Checkpoint 2 Passed
   2.
   Print `air_scatter` to view the plot. Consider whether this plot shows a sharp or fuzzy cutoff at 600 megawatts.
3. Checkpoint 3 Passed
   3.
   Add a line of code to `air_scatter` to add best-fit lines for each treatment group to the plot. Save the updated plot to `air_scatter2` .
4. Checkpoint 4 Passed
   4.
   Print `air_scatter2` . Do the regression lines appear to show a discontinuity in the outcome variable at the cutoff?

---

title: "Visual Check"

output: html-notebook

```{r}

# load ggplot2 library

library(ggplot2)

# import dataset

air_data <- read.csv("air_data.csv")

```

Checkpoint 1

```{r}
```

```
# create scatter plot with watts as forcing variable and AQI as outcome
air_scatter <- ggplot(
  data = air_data, #dataset
  aes(
    x = watts, #forcing variable
    y = aqi, #outcome variable
    color = group, #point color by treatment
    shape = group #point shape by treatment
  )
) +
  geom_point() + #add points to plot
  geom_vline(xintercept = 600, linetype = "dashed") #add dashed line
```

Checkpoint 2

```{r}
# print plot
print(air_scatter)
```

Checkpoint 3

```{r}
# add best-fit lines to air_scatter
air_scatter2 <- air_scatter +
  geom_smooth(aes(group = group), method = "lm")
```

Checkpoint 4

```{r, message=FALSE}
# print air_scatter2
print(air_scatter2)
```

Visual Check

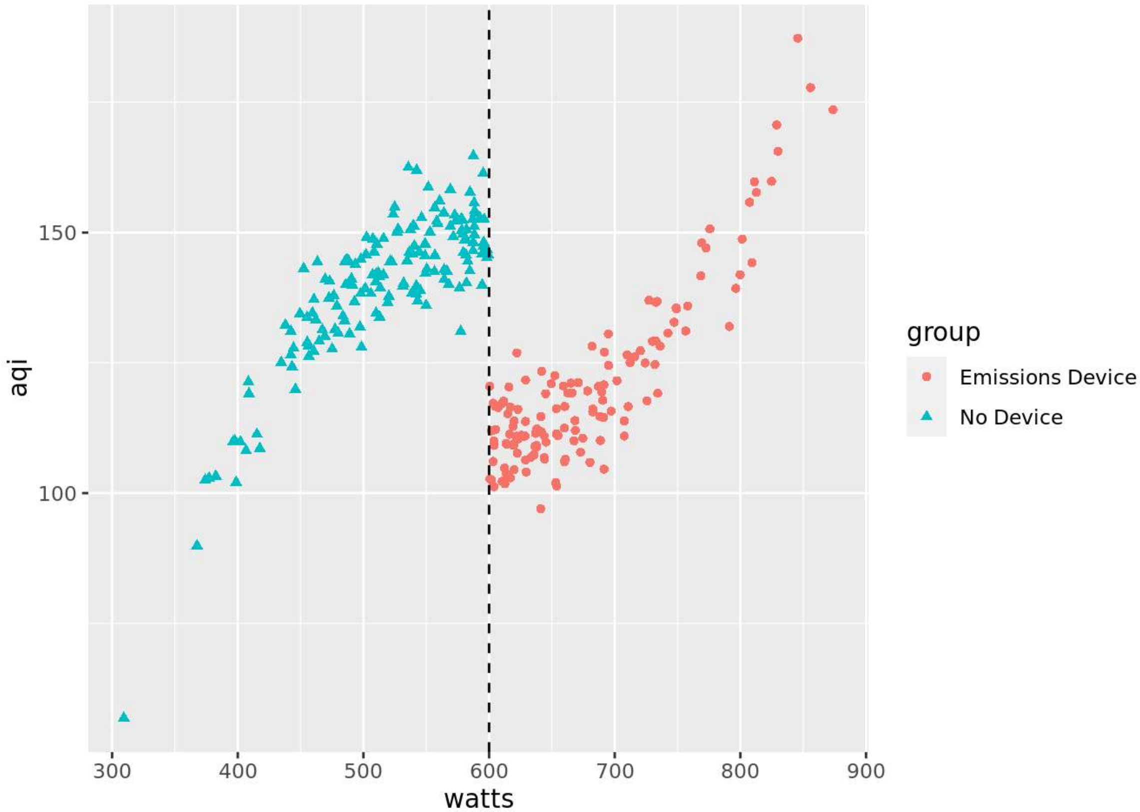
```
# load ggplot2 library
library(ggplot2)
# import dataset
air_data <- read.csv("air_data.csv")
```

Checkpoint 1

```
# create scatter plot with watts as forcing variable and AQI as outcome
air_scatter <- ggplot(
  data = air_data, #dataset
  aes(
    x = watts, #forcing variable
    y = aqi, #outcome variable
    color = group, #point color by treatment
    shape = group #point shape by treatment
  )
) +
  geom_point() + #add points to plot
  geom_vline(xintercept = 600, linetype = "dashed") #add dashed line
```

Checkpoint 2

```
# print plot
print(air_scatter)
```

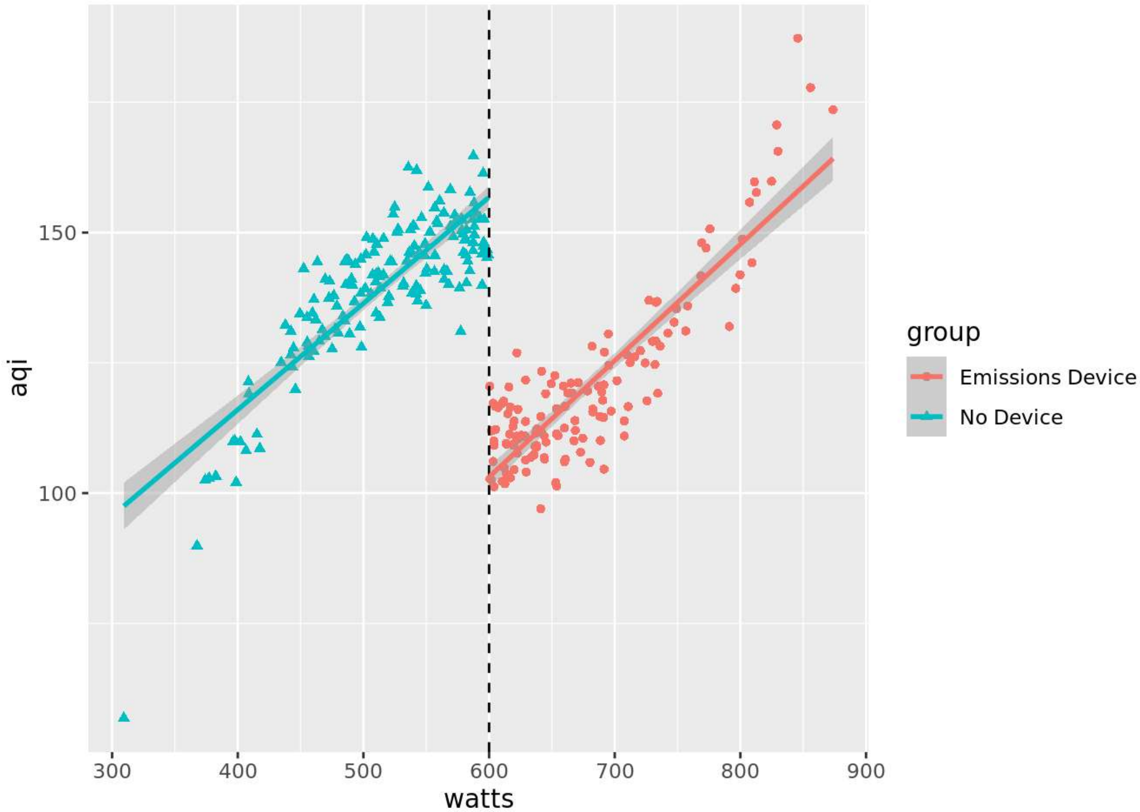


Checkpoint 3

```
# add best-fit lines to air_scatter
air_scatter2 <- air_scatter +
  geom_smooth(aes(group = group), method = "lm")
```

Checkpoint 4

```
# print air_scatter2
print(air_scatter2)
```



Choosing a Bandwidth

10 min

In RDD, we know we need to look at points near the cutoff to find treatment and control groups that are similar. But how do we know how close to look?

The *bandwidth* describes the distance on either side of the cutoff we should use to reduce our dataset. Any points that are more than one bandwidth above or below the cutoff are discarded. Choosing the bandwidth can have a serious impact on the results of an RDD analysis:

- A wider bandwidth keeps more of the original dataset, so we have more information to estimate the treatment effect with. However, the treatment groups might be too different on confounding
  variables
  Preview: Docs Loading link description
  , which could decrease accuracy.
- A narrower bandwidth retains less of the original dataset, so treatment groups will be more alike. However, the smaller sample size means less information to estimate the treatment effect.

We could select the bandwidth based on what we BELIEVE is best. However, an

algorithm

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that optimizes the bandwidth mathematically may be a better choice. A popular choice—which we will use—is the Imbens-Kalyanaraman (IK) algorithm.

The R package `rdd` contains all of the tools needed to calculate the optimal bandwidth and carry out an RDD analysis. To calculate the IK bandwidth using `rdd`, we will use the `IKbandwidth()` function, which requires three arguments:

- `X` : the forcing variable
- `Y` : the outcome variable
- `cutpoint` : the cutoff value to use.

To calculate the IK bandwidth for the contribution matching dataset, we would use the following code:

```
library(rdd)

# calculate IK bandwidth
cont_ik_bw <- IKbandwidth(
  X = cont_data$size, # forcing variable
  Y = cont_data$contribution, # outcome variable
  cutpoint = cont_cutpoint # cutpoint



)

# print the IK bandwidth to the console
cont_ik_bw
[1] 13.26322


Copy to Clipboard
```

The reduced dataset used in our RDD analysis will include only the companies that have between 286 and 314 employees (300 ± 13.26). Companies with between 286 and 314 employees are likely to be similar on other variables that may impact employee contributions, such as average salary or insurance costs.

To illustrate the bandwidth visually, we can add bandwidth lines to the scatterplot. We can use `geom_vline()` to add reference lines at the cutpoint ± the bandwidth to our scatter plot `rdd_scatter` from earlier:

```
rdd_scatter +
  geom_vline(xintercept = 300 + c(-cont_ik_bw, cont_ik_bw)) # add lines to
indicate the bandwidth


Copy to Clipboard
```

This plot shows us just how narrow the optimal bandwidth is for the contribution program dataset.

Instructions

1. Checkpoint 1 Passed
   1.
   The dataset `air_data` has been loaded for you in **notebook.Rmd**. Calculate the Imbens-Kalyanaraman (IK) optimal bandwidth using the `IKbandwidth()` function. Save the results to `air_ik_bw`.
2. Checkpoint 2 Passed
   2.
   Print `air_ik_bw`. Think about whether the bandwidth seems large or small in relation to the scale of the forcing variable.
3. Checkpoint 3 Passed
   3.
   A scatter plot of AQI (`aqi`) against power plant output (`watts`) with a dashed line at 600 megawatts has been created for you and saved as `air_scatter`. Modify `air_scatter` to add solid vertical lines for the bandwidth cutoff lines. Save the result to `air_scatter2`.
4. Checkpoint 4 Passed
   4.
   Print `air_scatter2`. Do you think the two device groups within this bandwidth will be similar enough to compare? What might be the tradeoff if the bandwidth is too narrow?

~~~~
title: "Choosing a Bandwidth"

output: html-notebook

---


```{r, message=FALSE}

# load rdd and ggplot2 libraries

library(rdd)

library(ggplot2)


# import data

air_data <- read.csv("air_data.csv")


# scatter plot with cutpoint

air_scatter <- ggplot(

  data = air_data, #dataset

  aes(

    x = watts, #forcing variable

    y = aqi, #outcome variable

    color = group, #color points by group

    shape = group #shape points by group

  )

) +

  geom_point() + #add points to plot

  geom_vline(xintercept = 600, linetype = "dashed") #add line

```

# Checkpoint 1

```{r}

# calculate the IK bandwidth

air_ik_bw <- IKbandwidth(

  X = air_data$watts, #forcing variable

  Y = air_data$aqi, #outcome variable

  cutpoint = 600 #cutpoint

)
~~~~

~~~
```

# Checkpoint 2

```{r}

# print air_ik_bw

print(air_ik_bw)

```

# Checkpoint 3

```{r}

# add vertical bandwidth lines to air_scatter

air_scatter2 <- air_scatter +

  geom_vline(xintercept = 600 + c(-air_ik_bw, air_ik_bw))

```

# Checkpoint 4

```{r}

# print updated plot

print(air_scatter2)

```
~~~

Choosing a Bandwidth

```r
# load rdd and ggplot2 libraries
library(rdd)
library(ggplot2)

# import data
air_data <- read.csv("air_data.csv")

# scatter plot with cutpoint
air_scatter <- ggplot(
  data = air_data, #dataset
  aes(
    x = watts, #forcing variable
    y = aqi, #outcome variable
    color = group, #color points by group
    shape = group #shape points by group
  )
) +
```

```
  geom_point() + #add points to plot
  geom_vline(xintercept = 600, linetype = "dashed") #add line
```

Checkpoint 1

```
# calculate the IK bandwidth
air_ik_bw <- IKbandwidth(
  X = air_data$watts, #forcing variable
  Y = air_data$aqi, #outcome variable
  cutpoint = 600 #cutpoint
)
```

Checkpoint 2

```
# print air_ik_bw
print(air_ik_bw)
```

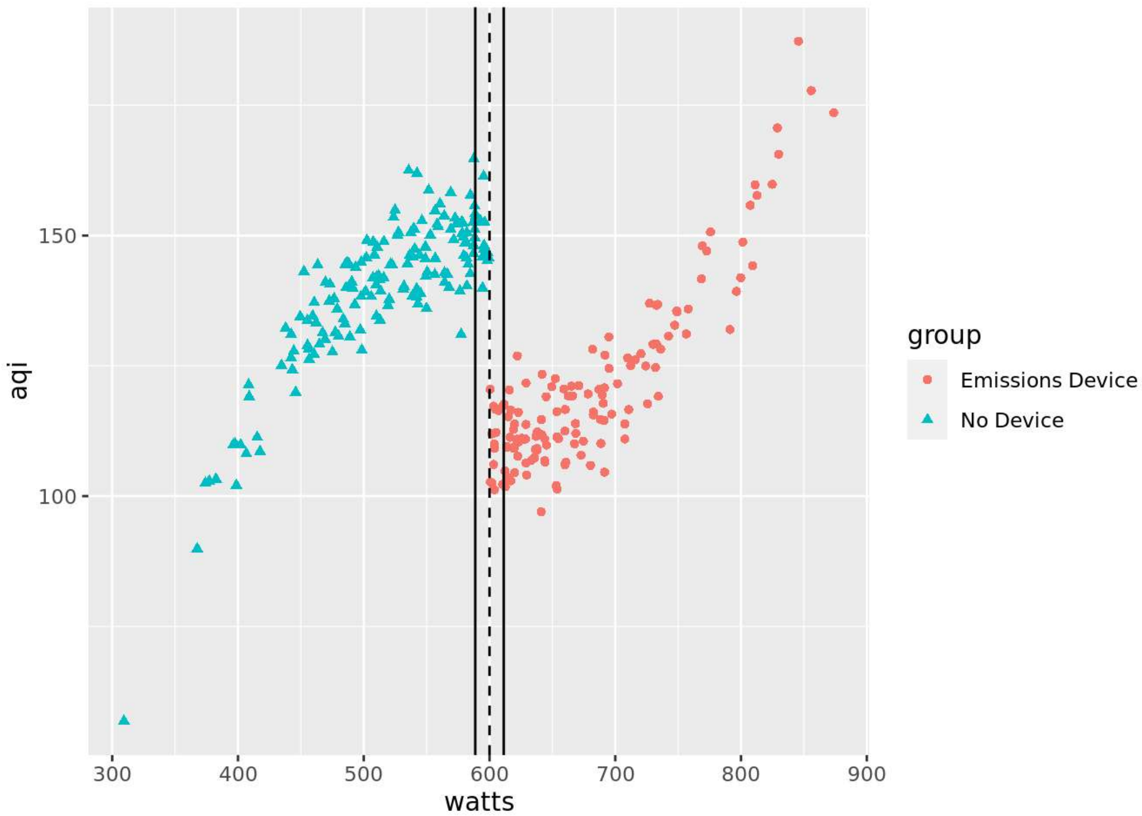
```
[1] 11.32436
```

Checkpoint 3

```
# add vertical bandwidth lines to air_scatter
air_scatter2 <- air_scatter +
  geom_vline(xintercept = 600 + c(-air_ik_bw, air_ik_bw))
```

Checkpoint 4

```
# print updated plot
print(air_scatter2)
```



Estimating the Causal Treatment Effect

7 min

The use of a bandwidth impacts the type of causal estimand we can calculate in a regression discontinuity design analysis. Because the RDD approach uses a subset of the full dataset, we can only estimate the *local average treatment effect* (LATE). The LATE is the average treatment effect among the subset of data that falls within the range of the bandwidth.

To estimate the LATE in RDD, a regression model that allows for different slopes on each side of the cutpoint is fit. The regression model is then used to get a predicted value of the outcome variable for each treatment group at the cutpoint. The difference between the predicted outcome values of the treatment and control groups is an estimate of the LATE.

We can use the `RDestimate()` function from the `rdd` package as follows to fit the local linear regression model for the contribution matching data:

```
cont_rdd <- RDestimate(
  formula = contribution ~ size, #outcome regression model
  data = cont_data, #dataset
  cutpoint = 300, #cutpoint
  bw = cont_ik_bw #bandwidth
)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```

The `RDestimate()` function fits the local linear regression model at the provided bandwidth, but also at half of the bandwidth and twice the bandwidth. If the estimate of the LATE is relatively the same across bandwidths, we can be more confident that the estimate is accurate. We see all three estimates when we print the results.

```
Call:
RDestimate(formula = contribution ~ size, data = cont_data,
cutpoint = 300, bw = cont_ik_bw)

Coefficients:
    LATE    Half-BW  Double-BW
   90.60     110.67   71.62


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```

The model output shows us that the LATE is 90.60, meaning that in this dataset, we can conclude that employer-sponsored retirement contribution matching programs led to an increase in average monthly contributions of $90.60. However, we see that the estimate changes based on the bandwidth, ranging from $110.67 at half of the bandwidth to $71.62 at twice the bandwidth.

Instructions

1. Checkpoint 1 Passed
   1.
   The `air_data` dataset has been loaded for you in **notebook.Rmd**. The IK optimal bandwidth has been saved as `air_ik_bw`. Use the `RDestimate()` function to fit the local linear regression model using `air_ik_bw` as the bandwidth. Save the results to `air_rdd`.
2. Checkpoint 2 Passed
   2.
   Print the results of the local linear regression. Take note of the estimates of the LATE at the different bandwidths. Are the estimates different or fairly similar? What do you think this says about the reliability of our estimate?

---

title: "Estimating the Causal Treatment Effect"

output: html-notebook

```{r, message=FALSE}

# load rdd and ggplot2 libraries

library(rdd)

library(ggplot2)

# import data
```

air_data <- read.csv("air_data.csv")

calculate the IK bandwidth

air_ik_bw <- IKbandwidth(

X = air_data$watts, #forcing variable

Y = air_data$aqi, #outcome variable

cutpoint = 600 #cutpoint

)

```

# Checkpoint 1

```{r}

# fit local linear regression model

air_rdd <- RDestimate(

formula = aqi ~ watts, #formula for outcome model

data = air_data, #dataset

cutpoint = 600, #cutpoint

bw = air_ik_bw #bandwidth

)

```

Checkpoint 2

```{r}

# print results of local linear regression

print(air_rdd)

```

Estimating the Causal Treatment Effect

```
# load rdd and ggplot2 libraries
library(rdd)
library(ggplot2)

# import data
air_data <- read.csv("air_data.csv")
```
```

```
# calculate the IK bandwidth
air_ik_bw <- IKbandwidth(
  X = air_data$watts, #forcing variable
  Y = air_data$aqi, #outcome variable
  cutpoint = 600 #cutpoint
)
```

Checkpoint 1

```
# fit local linear regression model
air_rdd <- RDestimate(
  formula = aqi ~ watts, #formula for outcome model
  data = air_data, #dataset
  cutpoint = 600, #cutpoint
  bw = air_ik_bw #bandwidth
)
```

Checkpoint 2

```
# print results of local linear regression
print(air_rdd)
```

```


Call:
RDestimate(formula = aqi ~ watts, data = air_data, cutpoint = 600,
    bw = air_ik_bw)

Coefficients:
     LATE    Half-BW  Double-BW
   -36.45     -32.82     -38.14









   Advantages and Disadvantages of RDD

6 min

As we've seen, the advantages of regression discontinuity design are that
RDD:

Is a simple
```

```
method
Preview: Docs A method is a small piece of code, usually defined in a class,
that can be used outside the class and in other parts of the program.
 to understand and implement.

Avoids using a complicated regression model for the entire dataset — the
local regression model is simple.

Is useful in cases where there is no overlap on a confounding variable,
which may prevent us from using stratification or propensity score analysis.

However, there are several drawbacks to RDD inherent to the method:

Smaller bandwidths make RDD assumptions more plausible BUT also reduce the
sample size.

The local average treatment effect (LATE) is not an easily interpretable
estimand. We calculated the effect of the contribution matching program only
among the companies with close to 300 employees. How confident can we be
that this effect would be the same in much smaller or larger companies?

Let’s explore the tradeoffs with an example. Say we run RDestimate() on the
contribution data again, but set the bandwidth to 100 instead of the IK
optimal bandwidth of 13. We save the results to rdd_100 and print the
output:

Call:
RDestimate(formula = contribution ~ size, data = cont_data,
cutpoint = 300, bw = 100)

Coefficients:
    LATE    Half-BW  Double-BW
   56.71      59.90      53.54


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We can also get information on the number of observations included and the
standard error of the LATE for each bandwidth with the code that follows.

rdd_100$obs #number of observations
# Output
[1] 113  68 178

rdd_100$se #standard errors
# Output
[1] 6.647 9.077 5.269
```

```
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Let’s consider just the half-bandwidth (50) and the double-bandwidth (200).
Using the half-bandwidth, we analyze only companies with 250-350 employees,
so we may believe these companies are very similar to one another. But this
leaves only 68 companies in our sample and a standard error of 9.077 for the
LATE. We may ask:

Can we trust a LATE with a higher standard error?

Do these findings apply to companies outside the range of 250-350 employees?

At the double-bandwidth, we analyze companies with 100-500 employees, so our
sample size is much larger at 178 companies and our standard error is
reduced to 5.269. But how confident are we that companies with 100 employees
are similar enough to compare to companies with 500 employees?

Regression discontinuity design is a useful method to keep in our causal
inference toolbox, but we must be aware of the tradeoffs throughout the
process.

Instructions

Checkpoint 1 Passed
1.
A new RDD model with a bandwidth of 50 called air_rdd_50 has been created
for you in notebook.Rmd. Print air_rdd_50 and inspect the results. Are there
big differences in the estimated LATE across bandwidths of 50, 25, and 100?

Checkpoint 2 Passed
2.
Check the sample size under each reported bandwidth of air_rdd_50. Is there
a big difference in sample size when you look at power plants with outputs
of 575-625 megawatts (Half-BW) and those with outputs of 500-600 megawatts
(Double-BW)?

Checkpoint 3 Passed
3.
Check the standard errors of the LATE under each reported bandwidth
of air_rdd_50. Is there a big difference in standard errors when you look at
power plants with outputs of 575-625 megawatts (Half-BW) and those with
outputs of 500-600 megawatts (Double-BW)? Does the standard error get larger
or smaller with bigger samples?

---
title: "Advantages and Disadvantages of RDD"
output: html-notebook
---
```

````
```{r, message=FALSE}
# load rdd and ggplot2 libraries
library(rdd)
library(ggplot2)

# import data
air_data <- read.csv("air_data.csv")

# fit model with bandwidth of 50
air_rdd_50 <- RDestimate(
  formula = aqi ~ watts, #outcome model
  data = air_data, #dataset
  cutpoint = 600, #cutpoint
  bw = 50 #bandwidth
)
```


# Checkpoint 1
```{r}
# Print model results
print(air_rdd_50)
```
# Checkpoint 2
```{r}
# Check the number of observations
air_rdd_50$obs
```


# Checkpoint 3
```{r}
# Check the standard errors
air_rdd_50$se
```


Advantages and Disadvantages of RDD

# load rdd and ggplot2 libraries
library(rdd)
library(ggplot2)

# import data
air_data <- read.csv("air_data.csv")

# fit model with bandwidth of 50
air_rdd_50 <- RDestimate(
  formula = aqi ~ watts, #outcome model
  data = air_data, #dataset
  cutpoint = 600, #cutpoint
  bw = 50 #bandwidth
````

```
)

Checkpoint 1

# Print model results
print(air_rdd_50)


Call:
RDestimate(formula = aqi ~ watts, data = air_data, cutpoint = 600,
    bw = 50)

Coefficients:
     LATE    Half-BW  Double-BW
   -39.37     -38.94     -40.56

Checkpoint 2

# Check the number of observations
air_rdd_50$obs

[1] 113  69 199

Checkpoint 3

# Check the standard errors
air_rdd_50$se

[1] 2.356950 3.122402 1.712759
```

Conclusion

1 min

In this lesson, we showed that the implementation of an employee-sponsored retirement matching program led to an increase in average monthly employee contributions.

We learned a lot about regression discontinuity design along the way, including:

- RDD is a
  method
  Preview: Docs A method is a small piece of code, usually defined in a class, that can be used outside the class and in other parts of the program.
  used when the treatment assignment is determined by a continuous forcing variable at a specific cutoff point.
- An RDD is known as *sharp* if the cutoff is exact and *fuzzy* if the cutoff is not exact.
- Individuals within a narrow window on either side of the cutoff are assumed to be similar to each other, except for the treatment group assignment.
- Local linear regression can be used to determine the local average treatment effect (LATE) among the individuals in this narrow window.
- Disadvantages of RDD include reduced sample size and potential lack of generalizability of the LATE.

Instructions

1. *Conclusion*
2. Great job performing a regression discontinuity design analysis on your own! In this lesson, you verified that RDD was an appropriate method to use, got causal estimates of the LATE using local linear regression, and examined the reliability of your results by using different bandwidths.
3. So what did this RDD analysis tell us? Based on the results of the local linear regression modeling (as shown in the learning environment), we can conclude that in our sample, emissions control devices led to a decrease in AQI of 36.45 points. This means that the emissions control device achieved its goal of providing cleaner air in close proximity to the power plants in our dataset.

Conclusion

Do emissions control devices decrease the AQI near power plants?

```
# Fit air quality model
RDestimate(
  formula = aqi ~ watts, #outcome model
  data = air_data, #dataset
  cutpoint = 600, #cutpoint
  bw = air_ik_bw #IK bandwidth
)
```

```


Call:
RDestimate(formula = aqi ~ watts, data = air_data, cutpoint = 600,
    bw = air_ik_bw)

Coefficients:
     LATE    Half-BW  Double-BW
```

-36.45 -32.82 -38.14

Instrumental

Variables

Introduction to Instrumental Variables

2 min

Suppose that a city is interested in increasing the per-capita rate of recycling among its citizens while decreasing the city’s operating costs.

To encourage this, the city allows individuals to discontinue curbside recycling pickup and instead opt into a rebate program. The city wants to evaluate whether or not this rebate program increases the amount of recycling in the city.

The city compiles the following

variables

Preview: Docs Loading link description

in a dataset to evaluate the success of the rebate program:

recycled: amount recycled (kg/person).
rebate: participation in rebate program (curbside vs. rebate).
distance: distance from recycling center (5 miles vs. > 5 miles). An individual who lives less than five miles from a recycling center might be more likely to opt into the rebate program than someone who lives more than five miles from a center. However, distance to a recycling center should not directly impact the amount of waste each person recycles.

We are going to use Instrumental Variables to answer the city’s question, but before we dive into that, we need to establish some more tools.

The first is conditional exchangeability.

One key assumption made in causal inference methods like weighting or stratification is conditional exchangeability. This assumption states that there are no unmeasured confounding variables that have a causal effect on both the treatment assignment and the outcome.

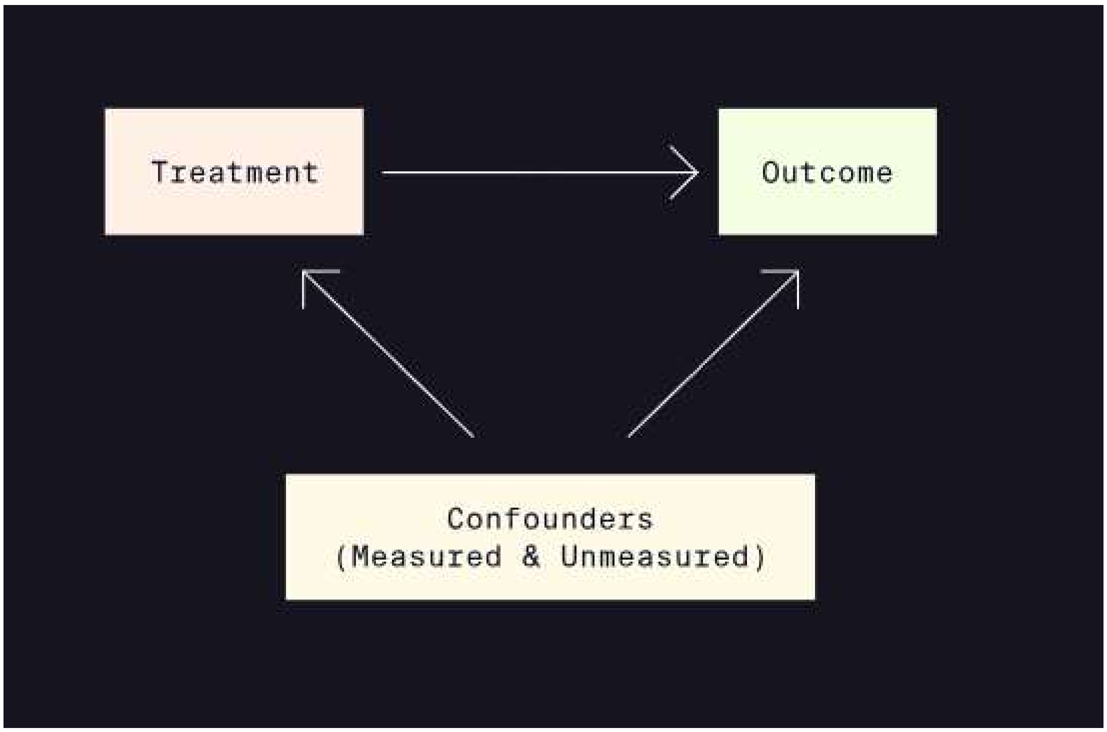
Randomization of the treatment assignment ensures that both measured and unmeasured confounding variables are evenly balanced between treatment groups. In a non-randomized setting, balance is NOT guaranteed. The

```
assumption of conditional exchangeability cannot be tested or verified — in
most cases, the best we can do without randomization is to identify and
measure as many potential confounding variables as possible.

Instrumental variable (IV) estimation is a causal inference technique that
helps us estimate the causal effect of the treatment even in the presence of
unmeasured confounding variables. In this lesson, we will learn about the
assumptions and potential applications of IV estimation.

Instructions

Take a look at the diagram in the learning environment. This diagram depicts
the causal relationships between the treatment, the outcome, and confounding
variables in a typical non-randomized study.
```



Not That Kind of Instrument

4 min

In observational studies, when randomization is not possible, balance of measured and unmeasured confounding

variables

Preview: Docs Loading link description

is not guaranteed. Without taking appropriate measures, the causal estimate of the effect of a treatment on an outcome of interest will be biased.

Instrumental variable (IV) estimation is one causal inference

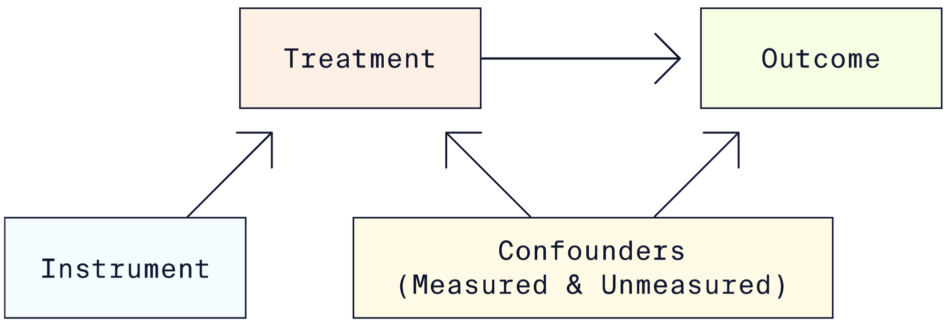
[method](#)

Preview: Docs Loading link description

that uses *instruments* to help reduce bias from both measured AND unmeasured confounding variables.

If you started this lesson hoping to learn about pianos and guitars, you may be disappointed. In IV estimation, an *instrument* (or *instrumental variable*) is a variable that is causally related to an outcome variable ONLY through another variable — typically the treatment variable of interest.

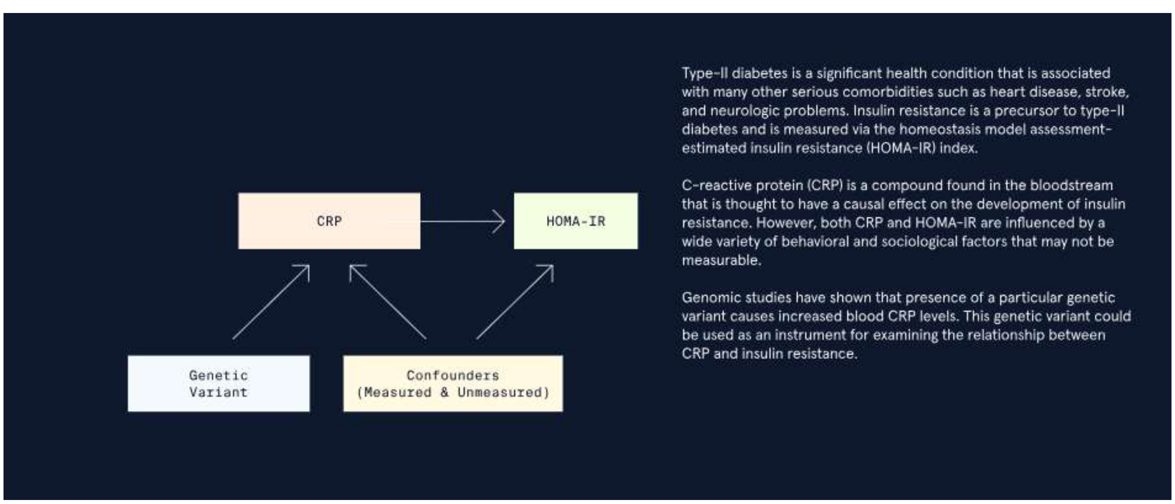
An instrumental variable would be depicted in a causal diagram as follows:

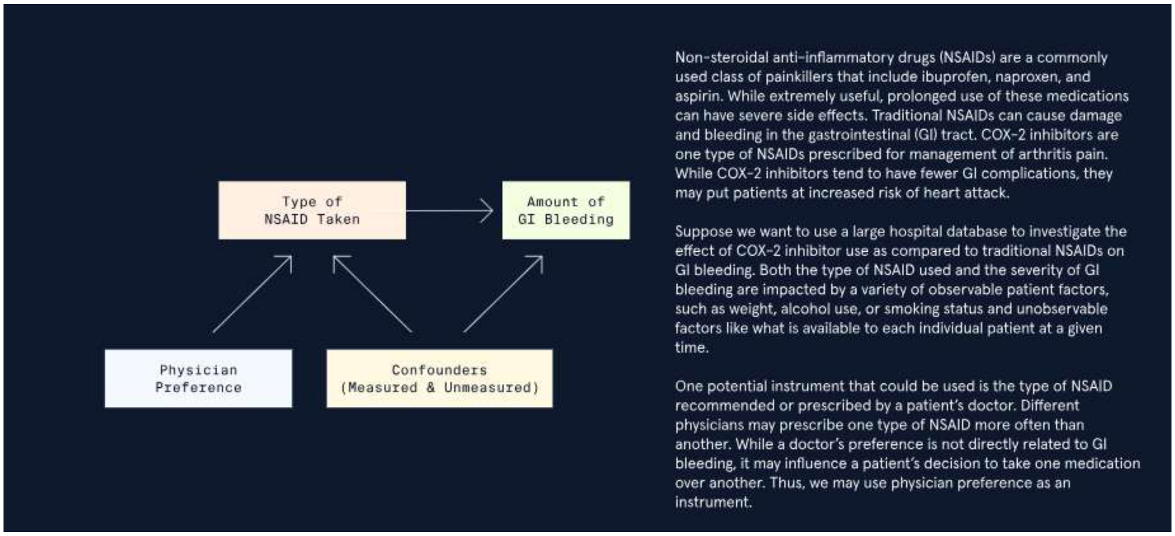


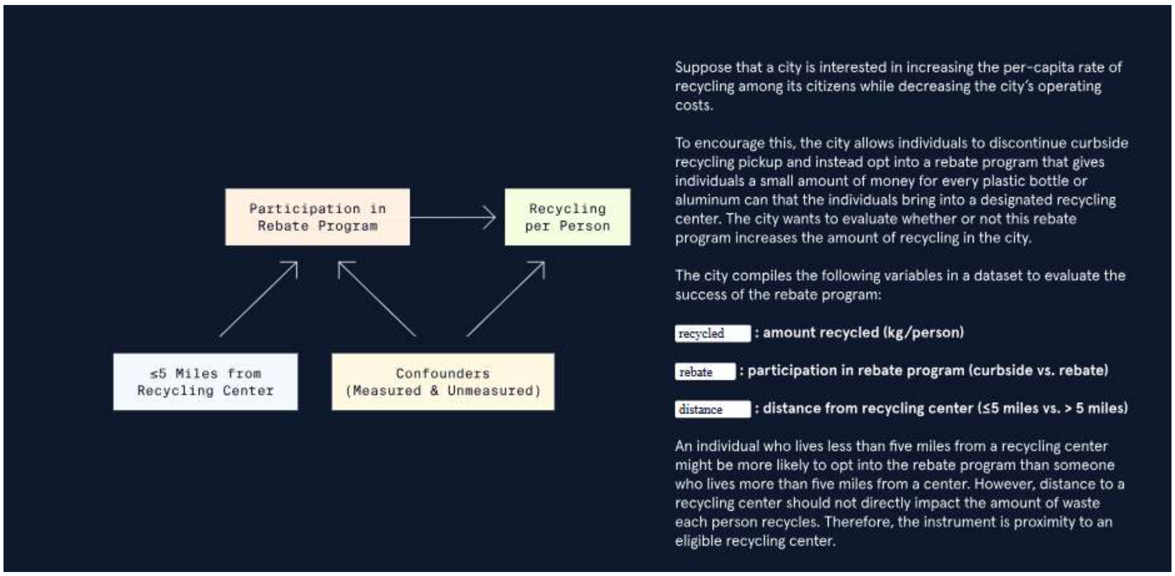
In this diagram, the arrows signify the presence and direction of a causal relationship. For example, there is no arrow directly from the instrument to the outcome because the instrument only impacts the outcome through its causal relationship with the treatment.

Instructions

1. Take a look at the animation in the learning environment. Each frame of the animation has a different example of how IV estimation could be used in practice along with a causal diagram that describes the scenario. Pause the animation as necessary to read and understand each scenario.
2. Pay particular attention to the third example, which describes the recycling scenario again. We will use it throughout the rest of this lesson to illustrate IV estimation.





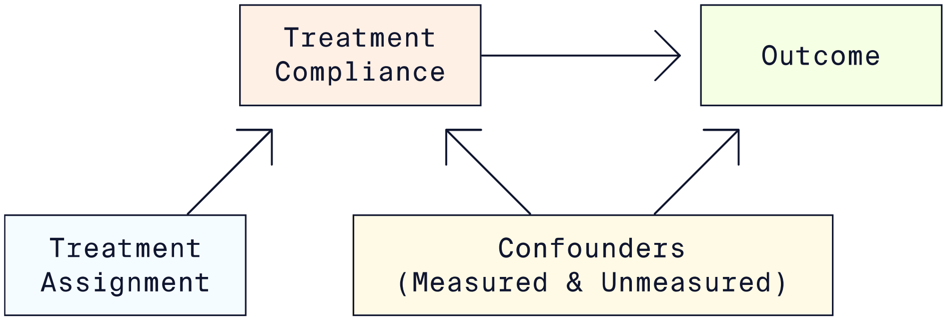


Assignment vs. Compliance

3 min

When treatment group assignments cannot be randomized due to ethical or practical reasons, the best we can do is to encourage compliance. However, encouragement does not guarantee compliance.

Compliance with the treatment assignment can be influenced by many factors, only some of which may be measurable. To account for unmeasured confounders of treatment compliance and the outcome, we could use IV estimation with treatment assignment as the instrument:



When the instrument AND treatment are both binary variables, we can define four types of "compliers":

1. *Always takers*: takes the treatment regardless of treatment assignment.
2. *Never takers*: never takes the treatment regardless of treatment assignment.
3. *Compliers*: takes the assigned treatment.
4. *Defiers*: takes the opposite of the assigned treatment.

In the context of the recycling program example, the four types of compliers would be defined as follows:

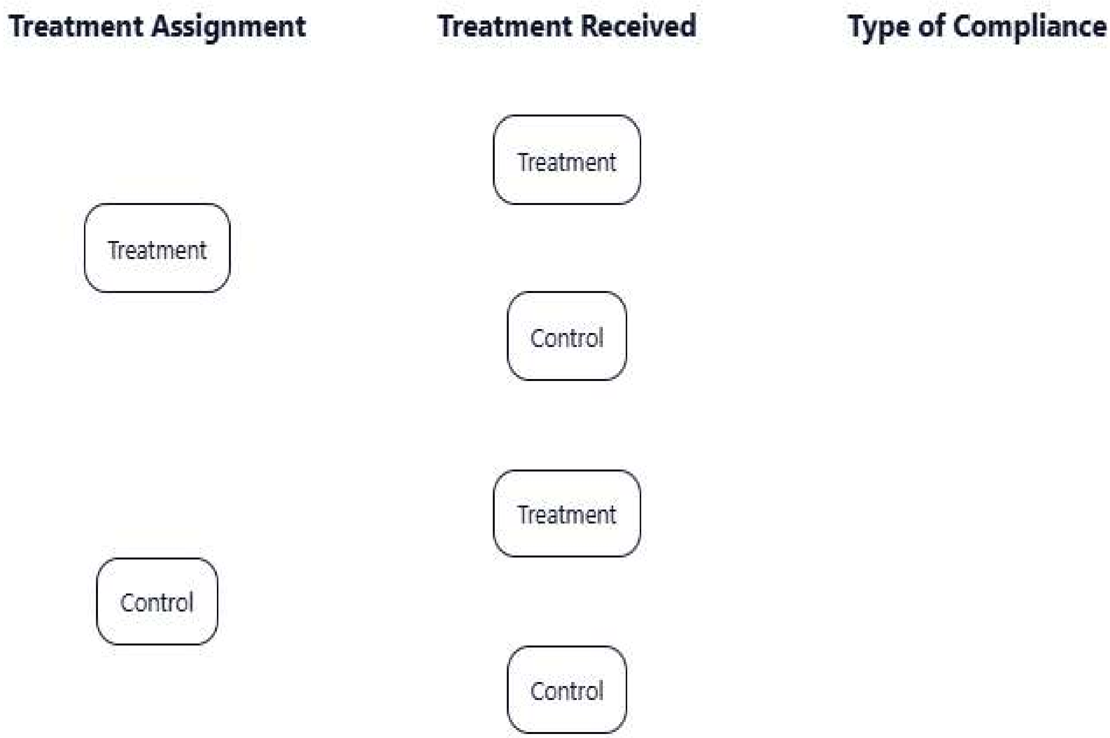
	Instrument Value	
	≤ 5 miles	**> 5 miles**
Always takers	Rebate	Rebate
Never takers	Curbside	Curbside
Compliers	Rebate	Curbside
Defiers	Curbside	Rebate

The issue of compliance is related to the fundamental problem of causal inference, which states that we can never observe both potential outcomes. In an IV estimation, we can never observe both potential treatments received for an individual. Thus, we cannot know with certainty what kind of complier an individual is.

Fortunately, we can get around this issue by making additional assumptions about our sample.

Instructions

1. Take a look at the interactive flowchart in the learning environment. The flowchart has three levels: treatment assignment, treatment received, and type of compliance. For each treatment assignment, click on one treatment received. Depending on your selection of treatment received, the type of compliance will appear.



Assumptions of IV Estimation

3 min

Compliance with the treatment assignment influences which causal estimand we calculate:

- When compliance is perfect, the treatment assignment and treatment received can be used interchangeably to get an accurate estimate of the causal effect.
- When compliance is NOT perfect, we cannot estimate the average treatment effect (ATE) because treatment assignment and treatment compliance are NOT interchangeable. Instead, the average effect of treatment assignment can be thought of as the effect of the *intention to treat*, or ITT effect.

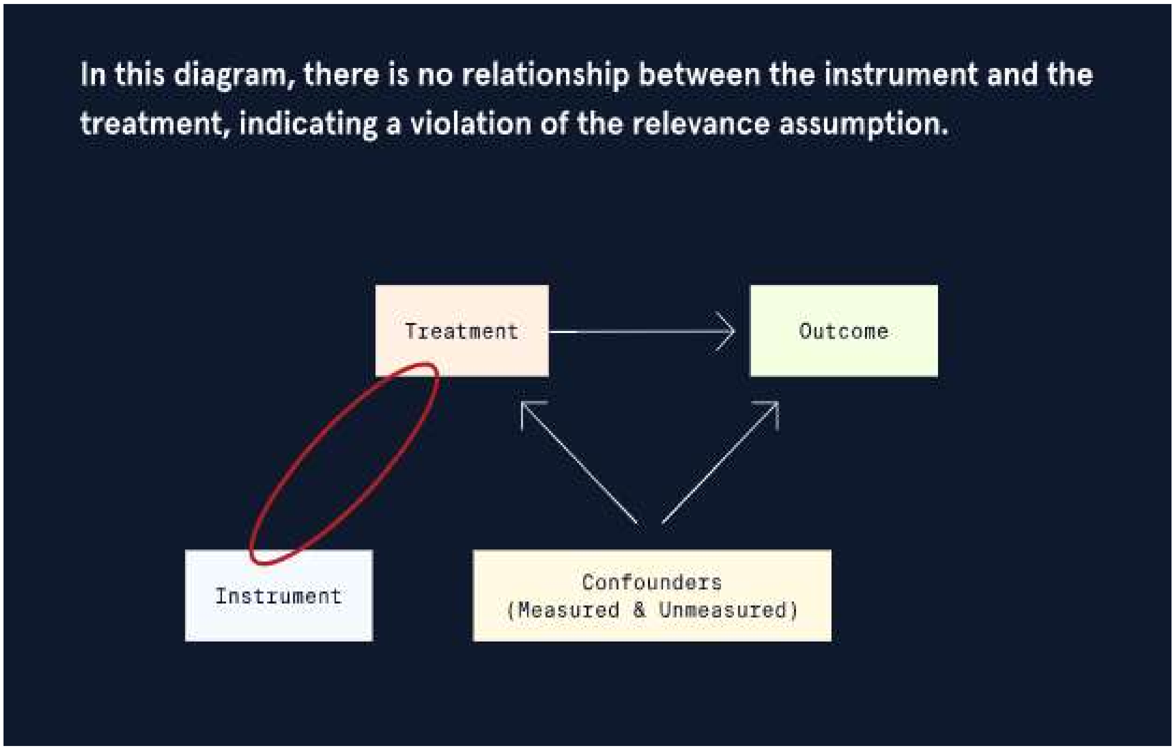
IV estimation allows us to approximate the ATE by estimating a local average treatment effect (LATE) just among the compliers. This is known as the *compliers' average causal effect* (CACE). To estimate the CACE, we must make four assumptions about the sample:

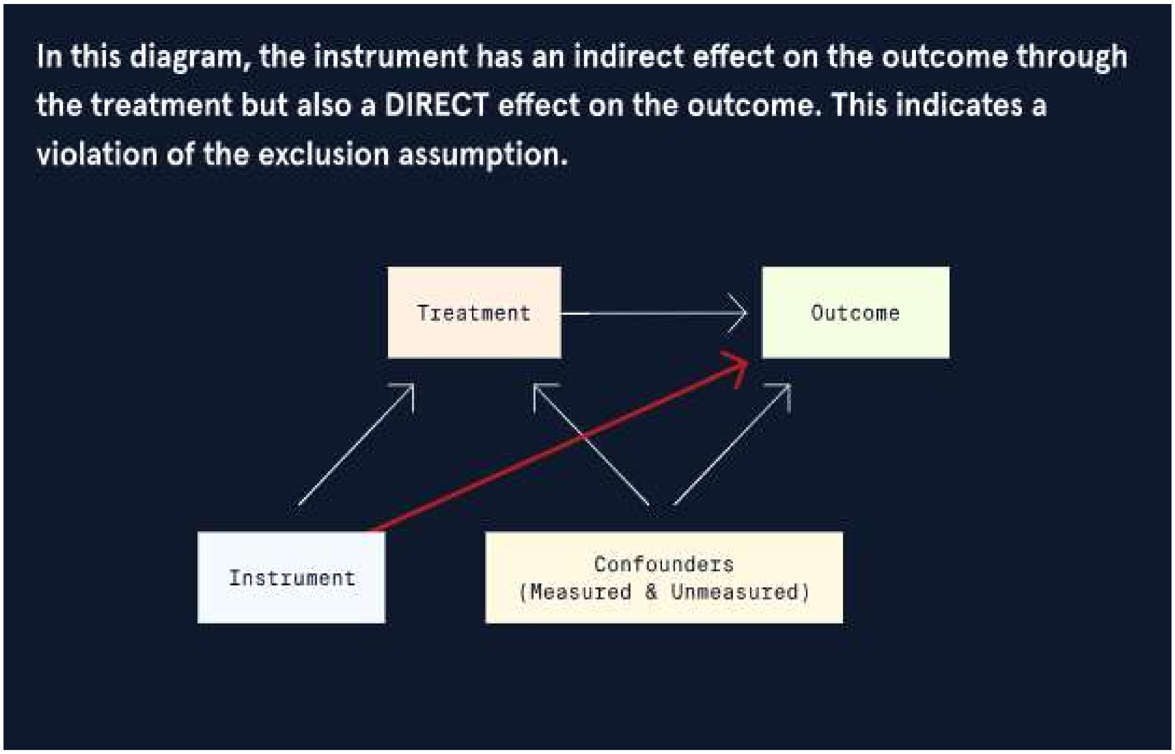
- *Relevance*: the instrument has a causal effect on the treatment received.
- *Exclusion*: the instrument affects the outcome ONLY indirectly through the treatment received.
- *Exchangeability*: there are no confounders that influence both the instrument AND the outcome.
- *Monotonicity*: there are no “defiers” in the sample.

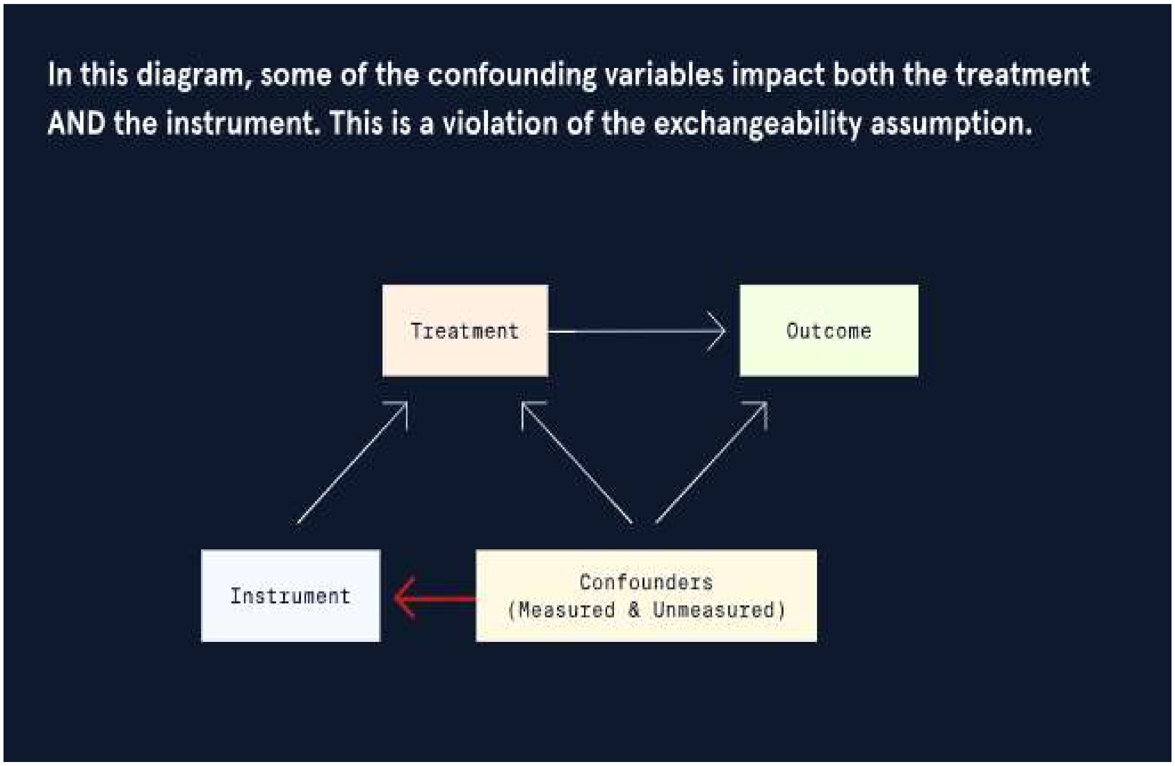
The relevance assumption implies that there are no “never-takers” and “always-takers” in the sample. In these subgroups, the treatment received doesn’t depend on the value of the assigned treatment. And in order to estimate the treatment effect among the “compliers,” we must also assume that there are no “defiers” in the sample.

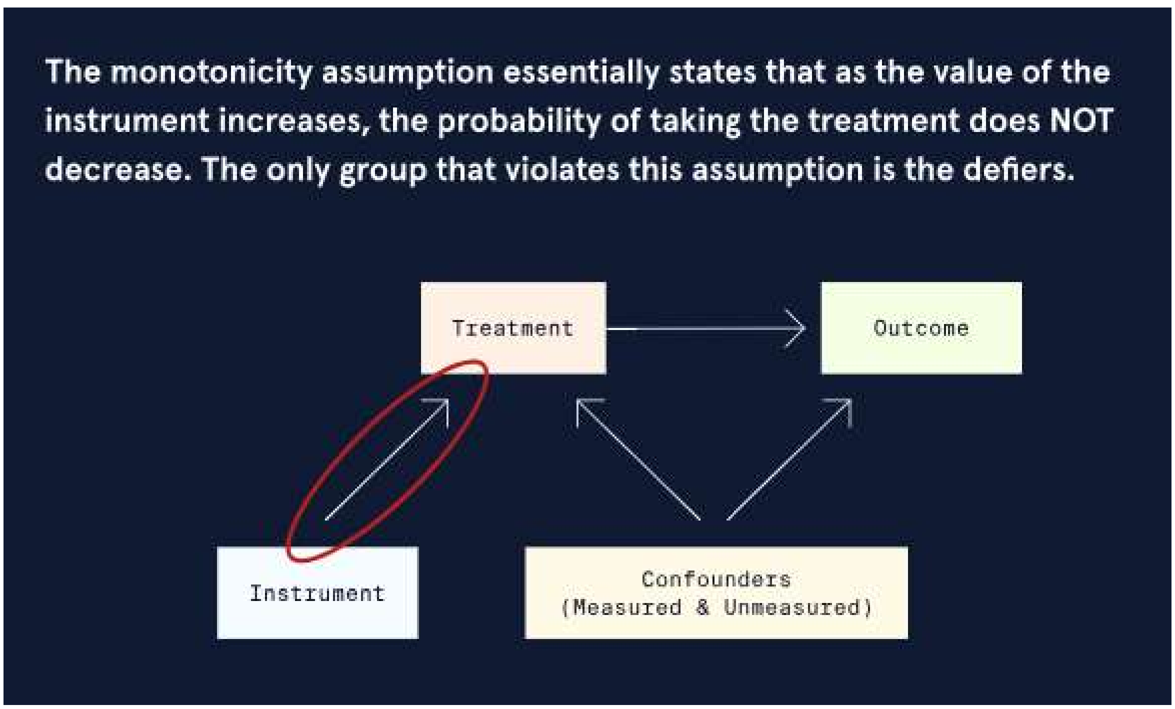
Instructions

1. The learning environment contains four pausable slides — one for each IV assumption. Check each slide for additional information about the assumptions, including diagrams showing violations of these assumptions.









Two-Stage Least Squares Regression

7 min

In a standard ordinary least squares (OLS) regression model, the outcome is predicted directly from the treatment assignment and other confounding variables. Such an OLS model would take the form:

Outcome = β_0 + β_1 * Treatment Assignment.

In OLS regression, we would use β_1 as an estimate of the treatment effect. To fit the OLS regression model using the recycling data, we would use the following code:

```
lm(recycled ~ rebate, #outcome ~ treatment
   data = recycle_df #dataset



)

# Output:

Call:
lm(formula = recycled ~ rebate, data = recycle_df)

Coefficients:
(Intercept)   rebate
  126.00       38.04


Copy to Clipboard
```

The OLS estimate suggests that participation in the rebate program leads to an average increase in recycling of 38.04 kilograms/person. However, this estimate is biased because OLS regression does not control bias introduced by unmeasured confounding variables or imperfect compliance.

In IV estimation, we account for unmeasured confounding variables and imperfect compliance via *two-stage least squares* (2SLS) regression. This type of regression predicts the outcome in two separate steps:

- In the first stage, treatment received is predicted by the instrument (treatment assignment): **Predicted Treatment Received = α_0 + α_1 * Treatment Assignment**
- In the second stage, the outcome is predicted as a function of the predicted treatment received from the first stage: **Outcome = β_0 + β_1 * Predicted Treatment Received**

β_1 from the second stage of the 2SLS regression model is used as the estimate of the CACE.

In this lesson, we focus on 2SLS regression with a continuous outcome, binary instrument, and binary treatment to keep things simple. When this is the case, the first stage uses logistic regression, while the second stage uses linear regression.

Instructions

1. Checkpoint 1 Passed
   1.
   An online retailer is interested in studying the effect of offering video streaming services on the average amount of money that users spend on the platform. The retailer gives its paying members access to exclusive video content at no additional expense. The online retailer then starts an email campaign for a random subset of users to encourage them to use the streaming services.
   The retailer tracks the amount of money spent by its users in a year and compiles the following variables into a dataframe named `video_df`:
   There are many factors that could influence whether or not an individual on the platform uses the video streaming services. For example, some individuals might have more free time to watch videos. Other individuals might not be interested in the content on the platform. The online retailer has no way to capture this information.
   Receiving the email does not directly impact how much someone spends — it only impacts the likelihood of using the streaming platform. Thus, the instrument is whether or not the individual was emailed.
   A file named **video_data.csv** has been provided which contains data collected by the online retailer. Import this file as a dataframe and name it `video_df`.
   - `email`: whether or not an individual received the email campaign (received email vs. did not receive email).
   - `streaming`: whether or not an individual used the streaming services (used service vs. did not use service).
   - `spending`: the amount of money spent by an individual in a year (dollars)

---

title: "Two-Stage Least Squares Regression"

output: html-notebook

Checkpoint 1

```{r}

# import data

video_df <- read.csv("video_data.csv")

```

Checkpoint 2

```{r}

# view structure of data

head(video_df)

```

Two-Stage Least Squares Regression

Checkpoint 1

```
# import data
video_df <- read.csv("video_data.csv")
```

Checkpoint 2

```
# view

  structure of data
head(video_df)
```

	streaming <int>	email <int>	spending <dbl>
1	1	1	277.1148
2	1	1	293.5193
3	0	0	249.3253
4	0	0	223.6334
5	1	0	299.3579
6	0	0	250.9672

6 rows

IV Estimation in R

9 min

Performing 2SLS in R is easy if we use the `ivreg()` function from the `AER` package.

The key difference in syntax between `ivreg()` and other regression functions is that the `formula`

argument

Preview: Docs Loading link description

of the `ivreg()` function must include the instrument. If we wanted to perform 2SLS regression with

[variables](#)

Preview: Docs Stores data that can be manipulated and referenced throughout the code.

`outcome` as the outcome, `treatment` as the treatment, and `instrument` as the instrument, the model formula would be `outcome ~ treatment | instrument`.

To fit the 2SLS regression using the recycling data, we would use the following code:

```
# import library
library(AER)

# run 2SLS regression
iv_mod <- ivreg(
  #outcome ~ treatment | instrument
  formula = recycled ~ rebate | distance,
  data = recycle_df
  )


Copy to Clipboard
```

To view the coefficients and standard errors, we can use `summary(iv_mod)$coefficients`, which gives the following output (you may need to make this section of the screen wider to view the full table):

```
             Estimate Std. Error   t value     Pr(>|t|)
(Intercept) 129.36463  0.8683141 148.98368 0.000000e+00
rebate       31.25452  1.4629239  21.36442 5.118885e-68




Copy to Clipboard
```

The results of 2SLS regression show that the estimate of the effect of the rebate program is 31.25, meaning participation in the rebate program led to an average increase in recycling of 31.25 kilograms/person. This only applies to compliers: those individuals who participated in the rebate program because they lived within 5 miles of a recycling center, but who would not have participated otherwise.

You may be wondering why we couldn't just fit the two separate regression models described in the previous exercise using `lm()` or `glm()` functions. The `ivreg()` function is preferred because it automatically corrects standard errors to account for the fact that the second stage regression model uses predicted values of the treatment.

If we use incorrect standard errors, we could make incorrect conclusions about the treatment effect:

- Lower standard errors correspond with more precise treatment effect estimates and a greater likelihood that the treatment coefficient will be found to be significantly different from zero.
- Higher standard errors correspond with less precise treatment effect estimates and a lesser likelihood that the treatment coefficient will be found to be significantly different from zero.

Instructions

1. Checkpoint 1 Passed
   1.
   Fit a linear ordinary least squares (OLS) regression model to estimate the effect of use of video streaming services on the amount spent by users of the online retailer. Save this regression model as `ols_model`.
2. Checkpoint 2 Passed
   2.
   Uncomment the code to print a summary of the coefficients from the ordinary least squares (OLS) model.
3. Checkpoint 3 Passed
   3.
   Use the `ivreg()` function from the AER package to fit the 2SLS regression model in one step. Make sure to modify the model formula to account for the instrument. Save this model as `iv_mod`.
4. Checkpoint 4 Passed
   4.
   Uncomment the `summary()` function to print the resulting coefficients. How does the estimate differ from the OLS estimate in the last checkpoint?

---

title: "IV Estimation in R"

output: html-notebook

```{r, message=FALSE}

# import library

library(AER)

# load dataset

video_df <- read.csv("video_data.csv")

```

Checkpoint 1

```{r}
```

~~~
# OLS model

ols_model <- lm(spending ~ streaming, data = video_df)

```

# Checkpoint 2

```{r}

# print OLS coefficients

summary(ols_model)

```

# Checkpoint 3

```{r}

# 2SLS model

iv_mod <- ivreg(

  #outcome ~ treatment | instrument

  formula = spending ~ streaming | email,

  data = video_df

)

```

# Checkpoint 4

```{r}

# print 2SLS model

summary(iv_mod)

```
~~~

IV Estimation in R

```r
# import library
library(AER)
# load dataset
video_df <- read.csv("video_data.csv")
```

Checkpoint 1

```r
# OLS model
ols_model <- lm(spending ~ streaming, data = video_df)
```

Checkpoint 2

```
# print OLS coefficients
summary(ols_model)
```

```


Call:
lm(formula = spending ~ streaming, data = video_df)

Residuals:
    Min      1Q  Median      3Q     Max 
-35.771  -8.262  -0.052   8.253  43.334 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 236.8673     0.6096  388.59   <2e-16 ***
streaming    45.0017     0.8838   50.92   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.68 on 698 degrees of freedom
Multiple R-squared:  0.7879,	Adjusted R-squared:  0.7876 
F-statistic:  2593 on 1 and 698 DF,  p-value: < 2.2e-16
```

Checkpoint 3

```
# 2SLS model
iv_mod <- ivreg(
  #outcome ~ treatment | instrument
  formula = spending ~ streaming | email,
  data = video_df
)
```

Checkpoint 4

```
# print 2SLS model
summary(iv_mod)
```

```


Call:
ivreg(formula = spending ~ streaming | email, data = video_df)

Residuals:
    Min      1Q  Median      3Q     Max 
```

```
-39.075  -8.389  -0.140   8.429  46.975

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  240.171      1.114  215.51   <2e-16 ***
streaming     38.056      2.133   17.84   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.18 on 698 degrees of freedom
Multiple R-Squared: 0.7691, Adjusted R-squared: 0.7688
Wald test: 318.2 on 1 and 698 DF,  p-value: < 2.2e-16
```

Interpretation and Considerations of IV Estimation

5 min

While IV estimation can be effective in certain circumstances, it also has limitations. One limitation is that the causal estimand (CACE) is not generalizable. The CACE only describes the effect of those who comply with treatment.

Another limitation is that it is difficult to find a suitable instrument that has a strong relationship with the treatment. If an instrument is only weakly related to the treatment, 2SLS regression will produce inaccurate estimates of the CACE.

To illustrate this, suppose that instead of using distance as an instrument for participation in the rebate program, we used another variable, `children`. The variable indicates whether or not an individual has children. Having children wouldn't directly cause a change in recycling, but individuals with children might be less likely to participate in the rebate program. The rebate program requires individuals to take the time to drop off recycling — time that people with children might not have.

Performing the 2SLS regression again with the `ivreg()` function and the `children` variable as an instrument highlights the effect of using a weak instrument:

```
iv_mod_weak <- ivreg(
  formula = recycled ~ rebate | children, #new weak instrument
  data = recycle_df
  )


Copy to Clipboard
```

Using `summary(iv_mod_weak)$coefficients` we can view just the coefficient table from the results summary.

```
             Estimate Std. Error   t value     Pr(>|t|)
(Intercept) 126.10140   8.930242 14.120715 5.477652e-37
rebate       37.84692  18.018181  2.100485 3.631487e-02
```

```
Copy to Clipboard
```

The estimate of 37.85 is neither accurate nor precise, as highlighted by the large standard error. This estimate is similar to the estimate from OLS regression, demonstrating that this is a weak instrument.

Instructions

1. Checkpoint 1 Passed
   1.
   Instead of using the email campaign as an instrument, let's see what happens when we choose a weaker instrument.
   Imagine that the online retailer decides to pay a search engine company to display an ad for the retailer's video streaming services above the results when anyone uses search terms like "video streaming" or "streaming". Seeing such an ad wouldn't directly increase the amount of money a user spends on the online retailer's website. However, the ad could make a user more likely to use the retailer's video streaming services. Thus, the targeted ads could act as an instrumental variable.
   This new variable, `ads`, can be found in a second dataframe called `video_df2` that has been loaded for you in **notebook.Rmd**.
   Fit a second 2SLS regression model that uses the `ads` variable as an instrument and save the results to `iv_mod2`.
2. Checkpoint 2 Passed
   2.
   The original 2SLS model with `email` as an instrument has been saved for you as `iv_mod`.
   Print a summary of the results of both 2SLS regression models saved in `iv_mod` and `iv_mod2`. How do the estimates of the treatment effect differ depending on what instrument is used? What about the standard errors?

---

title: "Interpretation and Considerations of IV Estimation"

output: html-notebook

```{r, message=FALSE}

library(AER) #load AER package

# import datasets

video_df <- read.csv("video_data.csv")

video_df2 <- read.csv("video_data2.csv")

# 2SLS model with email as instrument
```

iv_mod <- ivreg(

formula = spending ~ streaming | email,

data = video_df

)

```

# Checkpoint 1

```{r}

# 2SLS model with new instrument

iv_mod2 <- ivreg(

#use new instrument ads

formula = spending ~ streaming | ads,

data = video_df2

)

```

Checkpoint 2

```{r}

# Print summaries for both models

summary(iv_mod)

summary(iv_mod2)

```

Interpretation and Considerations of IV Estimation

```
library(AER) #load AER package

# import datasets
video_df <- read.csv("video_data.csv")
video_df2 <- read.csv("video_data2.csv")

# 2SLS model with email as instrument
iv_mod <- ivreg(
  formula = spending ~ streaming | email,
```
```

```
  data = video_df
)
```

Checkpoint 1

```
# 2SLS model with new instrument
iv_mod2 <- ivreg(
  #use new instrument ads
  formula = spending ~ streaming | ads,
  data = video_df2
)
```

Checkpoint 2

```
# Print summaries for both models
summary(iv_mod)
```

```

Call:
ivreg(formula = spending ~ streaming | email, data = video_df)

Residuals:
    Min      1Q  Median      3Q     Max 
-39.075  -8.389  -0.140   8.429  46.975 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  240.171      1.114  215.51   <2e-16 ***
streaming     38.056      2.133   17.84   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.18 on 698 degrees of freedom
Multiple R-Squared: 0.7691, Adjusted R-squared: 0.7688 
Wald test: 318.2 on 1 and 698 DF,  p-value: < 2.2e-16 
```

```
summary(iv_mod2)
```

```

Call:
ivreg(formula = spending ~ streaming | ads, data = video_df2)

Residuals:
    Min      1Q  Median      3Q     Max 
-47.629 -12.791  -0.389  12.663  56.402 
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   248.73      14.72  16.892   <2e-16 ***
streaming      20.08      30.92   0.649    0.516
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 17.08 on 698 degrees of freedom
Multiple R-Squared: 0.5462, Adjusted R-squared: 0.5455
Wald test: 0.4215 on 1 and 698 DF,  p-value: 0.5164
```

Conclusion

<1 min

Congratulations on finishing this lesson on instrumental variable estimation! We covered a lot of topics:

- Instrumental variable (IV) estimation is a causal inference technique that can be used to estimate a causal treatment effect even in the presence of unobserved confounding [variables](#)
  Preview: Docs Stores data that can be manipulated and referenced throughout the code.
  .
- IV estimation can be used in non-randomized studies when compliance with the assigned or encouraged treatment is not perfect.
- An instrument is a variable that is related to an outcome of interest ONLY through the treatment variable.
- The four assumptions of IV estimation are relevance, exclusion, exchangeability, and monotonicity.
- IV estimation is performed via two-stage least squares (2SLS) regression.
- The `ivreg()` function in the `AER` package performs 2SLS regression and automatically provides corrected standard errors of the treatment effect.

Instructions

1. *Conclusion:*
2. Great job! The 2SLS regression model using the email campaign as an instrument produced a CACE estimate of 38.056 (as shown in the learning environment). As long as our assumptions hold, this means that using video streaming services increased spending by an average of about 38 dollars. Since this is the CACE, it only applies to compliers: those who used the streaming services because they received the email, but who would not have used streaming otherwise.

Conclusion

```
library(AER)

video_df <- read.csv("video_data.csv")
```

```r
iv_mod <- ivreg(
  formula = spending ~ streaming | email, # full model with instrument on
right side
  data = video_df
)

summary(iv_mod)
```

```

Call:
ivreg(formula = spending ~ streaming | email, data = video_df)

Residuals:
    Min      1Q  Median      3Q     Max
-39.075  -8.389  -0.140   8.429  46.975

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  240.171      1.114  215.51   <2e-16 ***
streaming     38.056      2.133   17.84   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.18 on 698 degrees of freedom
Multiple R-Squared: 0.7691, Adjusted R-squared: 0.7688
Wald test: 318.2 on 1 and 698 DF,  p-value: < 2.2e-16
```

Based on the image, you're looking at a matching exercise for instrumental variable (IV) estimation assumptions. Two matches are correct (marked with green checks) and two are incorrect (marked with red X's). Here's the correct way to match them:

Correct Matching:

Assumption	Correct Definition
1. Relevance	The instrument has a causal effect on the treatme
2. Exchangeability	There are no confounding variables which influen instrument AND the outcome. ✅ (already correct
3. Exclusion	The instrument affects the outcome ONLY indirec treatment received.
4. Monotonicity	There are no "defiers" in the sample. ✅ (already

The Issue:

The two definitions were **swapped**:

- What was placed under **Relevance** actually belongs to **Exclusion**
- What was placed under **Exclusion** actually belongs to **Relevance**

Quick Explanation of Each Assumption:

1. **Relevance**: The instrument must actually affect the treatment — if there's no connection between instrument and treatment, IV won't work.
2. **Exchangeability**: The instrument shouldn't be correlated with unobserved factors that affect both treatment and outcome (like a randomized assignment).
3. **Exclusion Restriction**: The instrument can't directly change the outcome — it only works through its effect on the treatment.
4. **Monotonicity**: Everyone responds to the instrument in the same direction (no one would do the opposite of what the instrument suggests).

So the fix is to simply swap those two misplaced definitions!

Completed Code:

```r

# import library

library(ggplot2)


# RDD plot

ggplot(data = df, aes(x = forcing_var,

                      y = outcome_var,

                      color = trt_group,

                      shape = trt_group)) +

  geom_point()

```

Correct matches for the blanks:

- `x =` **forcing_var**

- `y =` **outcome_var**

- `color =` **trt_group**

- Layer: **geom_point()**

Why This Plot? (RDD Diagnostic Visualization)

In **Regression Discontinuity Design (RDD)**, the core idea is that treatment assignment is determined by a cutoff in the **forcing/running variable** (e.g., test score, age, income). We assume that individuals just above and below the cutoff are otherwise similar (continuity of potential outcomes).

This scatter plot is the **standard visual diagnostic** to:

1. Check for a **discontinuity** (jump) in the outcome at the cutoff → evidence of treatment effect.

2. Assess the functional form (linear? nonlinear?) on both sides.

3. Visualize how the treatment group (`trt_group`) behaves around the threshold.

Key assumptions this helps check:

- **Continuity of the forcing variable density** (no manipulation around cutoff).

- **Continuity of covariates** (no jumps in other variables).

- Functional form / bandwidth choice.

Enhanced Version (Recommended for Better Insight)

While the basic `geom_point()` completes the question, here's a more informative version used in practice:

```r
library(ggplot2)

ggplot(df, aes(x = forcing_var, y = outcome_var, color = trt_group)) +
  geom_point(alpha = 0.6, size = 2) +                    # Scatter
  geom_smooth(method = "lm", se = FALSE, linewidth = 1) + # Linear fit on each side
  geom_vline(xintercept = cutoff_value, linetype = "dashed", color = "red") +  # Cutoff line
```

```
  labs(title = "Regression Discontinuity Design - Visual Check",
       x = "Forcing Variable (e.g., Score)",
       y = "Outcome Variable",
       color = "Treatment Group") +
  theme_minimal()
```

Python equivalent (using `seaborn` + `matplotlib`, very common in Data Science):

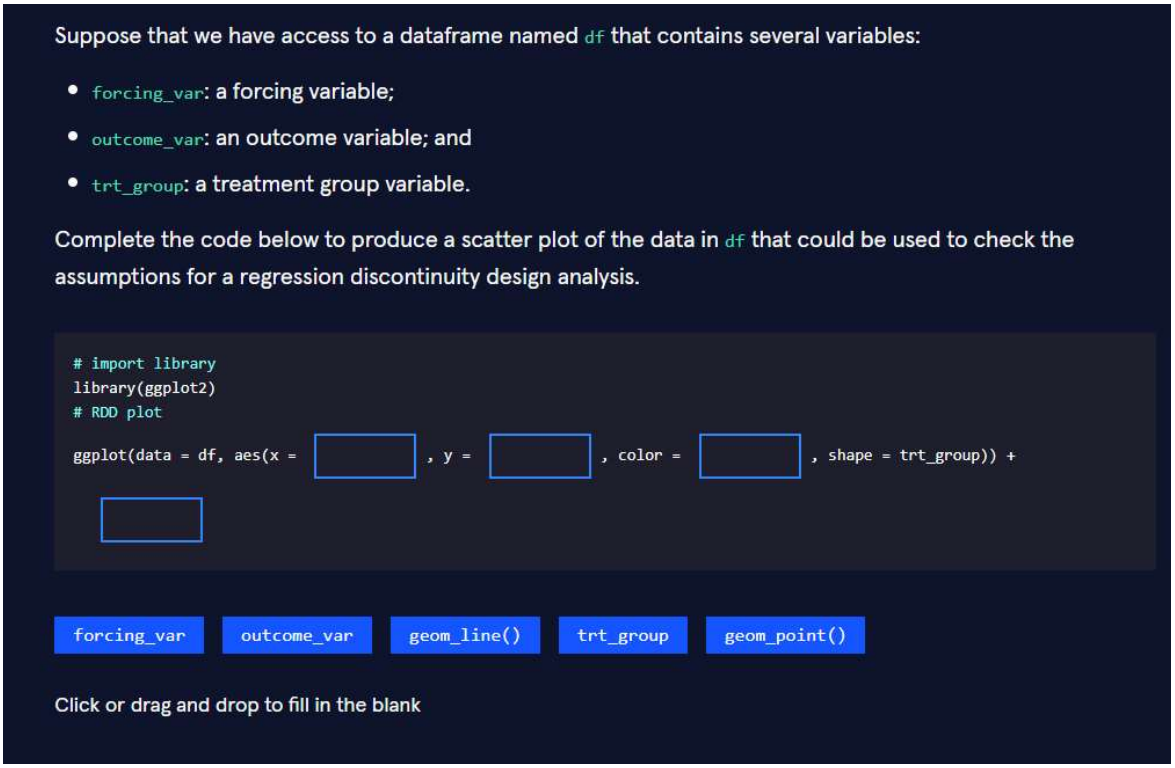
```python
import seaborn as sns
import matplotlib.pyplot as plt

sns.scatterplot(data=df, x='forcing_var', y='outcome_var',
                hue='trt_group', style='trt_group', alpha=0.7)
sns.lmplot(data=df, x='forcing_var', y='outcome_var',
           hue='trt_group', ci=None, scatter=False)
plt.axvline(x=cutoff_value, color='red', linestyle='--')
plt.title('RDD Scatter Plot for Assumption Checking')
plt.show()
```

Practical Guidance

- **When to use RDD**: Sharp cutoff rules (e.g., scholarship if score > 80, policy implemented at a date, etc.).
- **Next steps after this plot**:
  1. McCrary density test (check manipulation).
  2. Formal RDD estimation (`rdrobust` package in R, or `rdd` / `rd` in Python).
  3. Robustness checks with different bandwidths and polynomial orders.

Would you like the full `rdrobust` code example, a simulated dataset to test this on, or help interpreting what a "good" vs. "bad" RDD plot looks like?



Suppose that we have access to a dataframe named `df` that contains several variables:

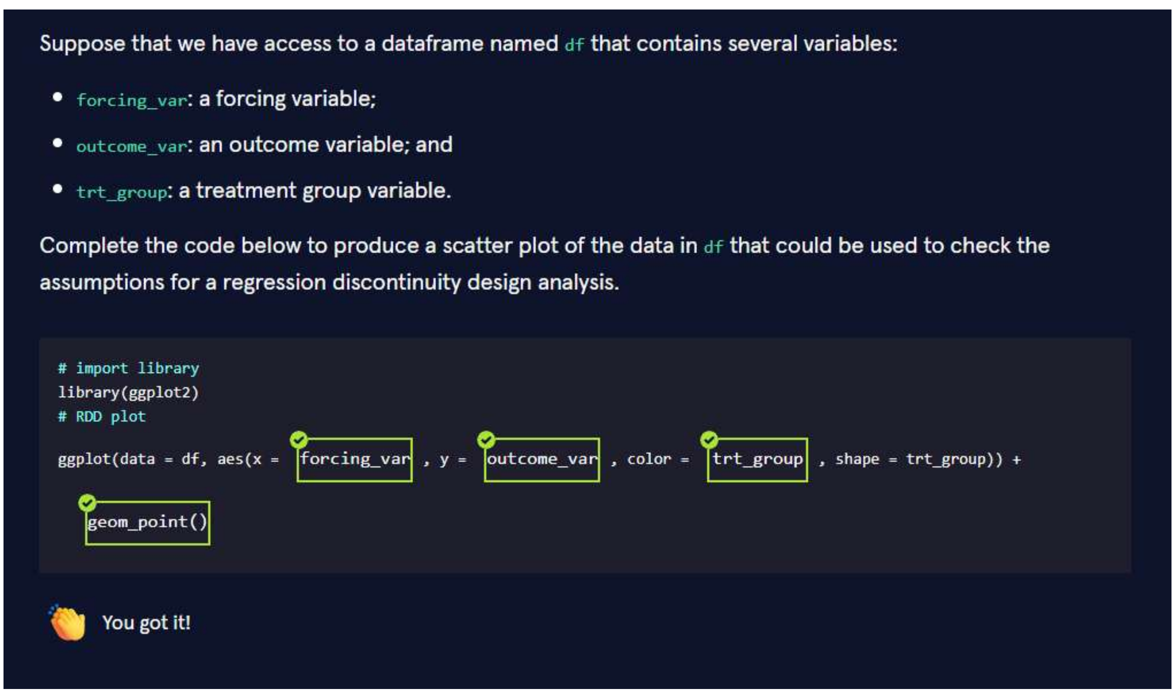
- `forcing_var`: a forcing variable;
- `outcome_var`: an outcome variable; and
- `trt_group`: a treatment group variable.

Complete the code below to produce a scatter plot of the data in `df` that could be used to check the assumptions for a regression discontinuity design analysis.

```
# import library
library(ggplot2)
# RDD plot
ggplot(data = df, aes(x = [ ] , y = [ ] , color = [ ] , shape = trt_group)) +
  [ ]
```

`forcing_var` `outcome_var` `geom_line()` `trt_group` `geom_point()`

Click or drag and drop to fill in the blank



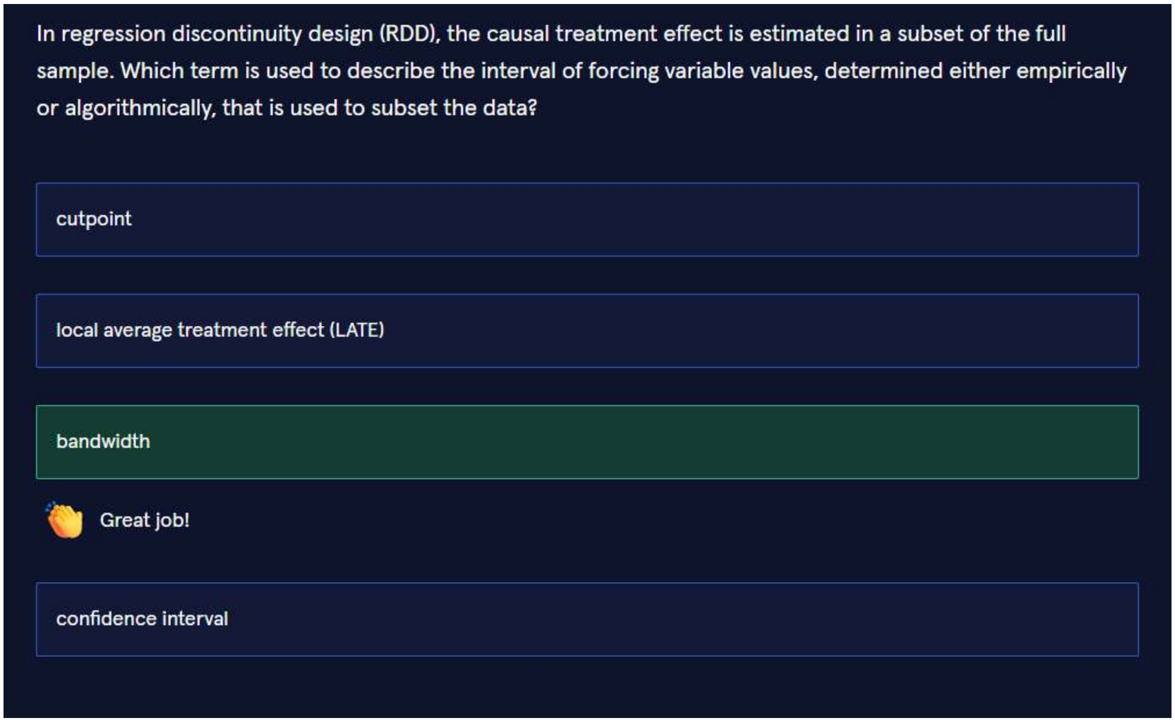
Suppose that we have access to a dataframe named `df` that contains several variables:

- `forcing_var`: a forcing variable;
- `outcome_var`: an outcome variable; and
- `trt_group`: a treatment group variable.

Complete the code below to produce a scatter plot of the data in `df` that could be used to check the assumptions for a regression discontinuity design analysis.

```
# import library
library(ggplot2)
# RDD plot
ggplot(data = df, aes(x = forcing_var , y = outcome_var , color = trt_group , shape = trt_group)) +
  geom_point()
```

You got it!



Correct Answer: bandwidth

Detailed Explanation

In **Regression Discontinuity Design (RDD)**, we exploit a cutoff (or cutpoint) in a continuous **forcing variable** (also called running variable) to estimate causal effects. However, we rarely use the entire dataset. Instead, we focus on observations **close to the cutoff** — this is where the key identifying assumption (continuity of potential outcomes) is most plausible.

The term for this interval/window around the cutoff is called the **bandwidth** (often denoted as \(h\)).

- **Bandwidth** = the width of the window on each side of the cutoff.
  - Example: If the cutoff is 0 and bandwidth \(h = 10\), we use observations where forcing variable is in \([-10, +10]\).

This bandwidth can be chosen:
- **Empirically** (researcher picks based on domain knowledge).
- **Algorithmically** (data-driven methods like cross-validation, IK bandwidth selector, etc.).

Visual Representation

```
Forcing Variable
<-------------------|------------------->
   Left side        Cutoff         Right side
   (control)         |              (treated)
          <-------- Bandwidth (h) -------->
```

Only data points inside this bandwidth are used to estimate the jump (discontinuity) at the cutoff.

Why Not the Other Options?

Term	Meaning	Why Not Correct Here?
cutpoint	The threshold value itself (e.g., score = 80)	This is the point, not the interval around it
local average treatment effect (LATE)	The causal parameter we estimate for units near the cutoff	This is the *result*, not the data subset
bandwidth	The interval/window around the cutoff used to subset the data	**Correct**
confidence interval	Statistical uncertainty around the estimated effect	This is for inference, not data selection

---

Practical Code Examples

R (using `rdrobust` package — gold standard for RDD):

```r
library(rdrobust)

# Automatic bandwidth selection
rdd_result <- rdrobust(y = outcome_var,
                       x = forcing_var,
                       c = cutoff_value)   # c = cutpoint

summary(rdd_result)   # Shows optimal bandwidth used
```

Python (using `rdd` or manual subsetting):

```python
import numpy as np
from sklearn.linear_model import LinearRegression

# Manual bandwidth subset
bandwidth = 10.0   # or use data-driven selection
subset = df[np.abs(df['forcing_var'] - cutoff) <= bandwidth]

# Local linear regression on each side
left = subset[subset['forcing_var'] < cutoff]
right = subset[subset['forcing_var'] >= cutoff]

model_left = LinearRegression().fit(left[['forcing_var']], left['outcome_var'])
model_right = LinearRegression().fit(right[['forcing_var']], right['outcome_var'])

effect = model_right.predict([[cutoff]])[0] - model_left.predict([[cutoff]])[0]
print("Estimated RDD effect:", effect)
```

Guidance for Real-World Use

- **Small bandwidth** → more credible local comparison, but higher variance (fewer observations).

- **Large bandwidth** → more precision, but risk of bias if the relationship is nonlinear far from cutoff.

- **Best practice**: Always report results for multiple bandwidths + use robust bias-corrected estimators (like in `rdrobust`).

Would you like a full worked example with simulated data, code to visualize the bandwidth window, or help choosing bandwidth selectors?

Which of the following is NOT one of the key assumptions for performing a sharp regression disconinuity design analysis?

Treatment impacts the outcome but is independent of other variables.

There can be several cutpoints that determine the treatment group assignment.

Perfect! In a sharp RDD analysis, we assume that there is only one cutpoint that determines the treatment group assignment.

Both counterfactual outcomes can be modeled within a narrow interval around the cutpoint.

Within a narrow interval around the cutpoint, the treatment groups are exchangeable.

Which of the following is NOT true about regression discontinuity design (RDD)?

When a forcing variable cutpoint is exact, the RDD is known as a sharp RDD. When the cutpoint is not exact, the RDD is known as a fuzzy RDD.

In RDD, a forcing variable is a continuous variable with a cutpoint that determines the treatment group assignment.

In RDD, the forcing variable is NOT related to the outcome variable of interest.

👏 Correct! In regression discontinuity design, the forcing variable determines the treatment assignment AND is also related the outcome variable.

RDD may be used for causal inference when the treatment is assigned to observations that fall above or below a specific cutpoint.

Which of the following accurately describes the purpose and stages of two-stage least squares (2SLS) regression for instrumental variable (IV) estimation?

2SLS is used to estimate the causal treatment effect in IV estimation. The first stage predicts the instrument from the treatment received. The second stage uses the instrument predicted in the first stage to predict the outcome.

2SLS is used to estimate the causal treatment effect in IV estimation. The first stage predicts the treatment received from the instrument. The second stage uses the predicted treatment received in the first stage to predict the outcome.

👏 Correct!

2SLS is used to estimate the instrument effect in IV estimation. The first stage predicts the outcome from the treatment received. The second state uses the predicted outcome to estimate the instrument.

True or False: In instrumental variable estimation, the intention-to-treat (ITT) effect is ALWAYS the same as the compliers' average causal effect (CACE), which is a type of local average treatment effect (LATE).

True

False

Great job! The ITT effect measures the effect of the treatment assignment, whereas the CACE measures the effect of the treatment received. These are only interchangeable with perfect compliance.

True or False: the primary disadvantages of instrumental variable (IV) estimation are that the causal estimand is not generalizable and that finding a good instrument is difficult.

False

True

Correct!

Which of the following is NOT true about instrumental variable (IV) estimation?

Instrumental variable estimation produces accurate estimates of the treatment effect even when the instrument is weakly related to the treatment.

Correct! Weak instruments generally produce biased estimates of the treatment effect.

In IV estimation, treatment assignment and the treatment actually received are treated as separate factors.

In IV estimation, an instrument is a variable that is related to an outcome indirectly through another variable — typically the treatment variable.

Instrumental variable estimation helps account for bias even in the presence of unmeasured confounding variables.

Correct Answer:

The first statement is NOT true:

> "Instrumental variable estimation produces accurate estimates of the treatment effect even when the instrument is weakly related to the treatment."

Why This Statement is False

A **weak instrument** (one that is only weakly related to the endogenous treatment variable) is one of the **biggest problems** in IV estimation.

Consequences of weak instruments:

- Severe finite-sample bias (can be worse than OLS).

- Inflated variance and unreliable confidence intervals.

- The IV estimator becomes highly sensitive to small violations of other assumptions.

- Estimates can be inconsistent.

Rule of thumb: The first-stage F-statistic (strength of the instrument) should be **greater than 10**. Many researchers now use much stricter thresholds.

Why the Other Statements Are True

Statement	Status	Explanation
Treatment assignment and treatment received are treated as separate	**True**	Especially in **fuzzy IV** designs, we distinguish the instrument (assignment) from actual treatment received.
Instrument is related to outcome indirectly through treatment	**True**	This is the **exclusion restriction** — the core of how IV works (Z → X → Y).
Helps account for bias from unmeasured confounding	**True**	This is the main purpose of IV: it uses exogenous variation from the instrument to isolate causal effects despite unmeasured confounders.

Practical Diagnostics & Code

R Code (Check Instrument Strength)

```r
library(AER)

first_stage <- lm(treatment ~ instrument + controls, data = df)
summary(first_stage)   # Look at F-statistic for the instrument

# Full 2SLS
iv_model <- ivreg(outcome ~ treatment + controls | instrument + controls, data = df)
summary(iv_model, diagnostics = TRUE)
```

Python Code

```python
from linearmodels.iv import IV2SLS
import statsmodels.api as sm

# First stage for diagnostics
first_stage = sm.OLS(df['treatment'],
                     sm.add_constant(df[['instrument', 'controls']])).fit()
print("First-stage F-stat:", first_stage.fvalue)

# 2SLS model
model = IV2SLS(df['outcome'], sm.add_constant(df[['controls']]),
               df[['treatment']], df[['instrument']]).fit()
print(model)
```

Summary of IV Trade-offs

Advantages: Can handle unmeasured confounding.

Disadvantages:

- Requires strong, valid instruments (hard to find).

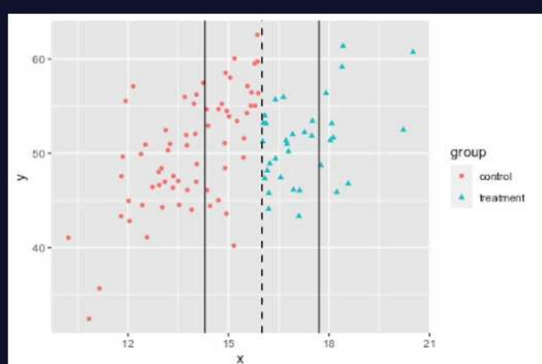
- Estimates **Local Average Treatment Effect (LATE)**, not ATE.

- Weak instruments can be disastrous.

Would you like examples of **weak vs strong instruments**, a full simulation showing bias from weak instruments, or help evaluating a specific instrument you have in mind?

The scatter plot shown represents a dataset named `df` with a forcing variable `x` and an outcome variable `y`. A third variable, `group`, represents the treatment group. Assume that the `group` variable is based on using a cutpoint value of 16. The optimal bandwidth is calculated to be 1.7.

Which of the following statements describes the correct model for regression discontinuity design AND has the correct code for the model?



The local average treatment effect (LATE) is estimated by fitting a local regression model that allows for different slopes on either side of the cutpoint.

```
# import library
library(rdd)
# model the LATE
RDestimate(
  formula = y ~ x,
  data = df,
  cutpoint = 16,
  bw = 1.7
)
```

Great job! This would fit the specified linear regression model.

The local average treatment effect (LATE) is estimated by fitting a local regression model that allows for different slopes on either side of the cutpoint.

```
# import library
library(rdd)
# model the LATE
RDestimate(
  formula = y ~ x,
  data = df,
  cutpoint = 1.7,
  bw = 16
)
```

The local average treatment effect (LATE) is estimated by fitting a linear regression of outcome predicted by the treatment group and forcing variable.

```
lm(y ~ group + x, data = df)
```

The local average treatment effect (LATE) is estimated by fitting a simple linear regression of outcome predicted by the forcing variable.

```
lm(y ~ x, data = df)
```

Correct Answer:

The first option (top box):

> "The local average treatment effect (LATE) is estimated by fitting a local regression model that allows for different slopes on either side of the cutpoint."

Why This is Correct

In **Regression Discontinuity Design (RDD)**, the standard way to estimate the **Local Average Treatment Effect (LATE)** is to fit a **local regression** (or local polynomial) **within the optimal bandwidth** around the cutoff, **allowing different slopes** on the left and right sides.

This matches:

- Cutoff = **16**

- Optimal bandwidth = **1.7**

The model should capture:

- A possible **jump** (discontinuity) at the cutoff.

- Potentially **different trends** (slopes) on each side.

Correct Code Approaches

Best R code (using `rdrobust` — recommended package):

```r
library(rdrobust)

rd_model <- rdrobust(y = df$y,
                     x = df$x,
                     c = 16,        # cutoff
                     h = 1.7)       # bandwidth

summary(rd_model)
```

```

**Using the `rdd` package (as shown in the question):**

```r
library(rdd)

rd_estimate <- RDestimate(y ~ x,
                          data = df,
                          cutpoint = 16,
                          bw = 1.7)
summary(rd_estimate)
```

Manual local linear regression with interaction (transparent approach):

```r
# Within bandwidth
df_local <- df[abs(df$x - 16) <= 1.7, ]

model <- lm(y ~ group * x, data = df_local)   # group * x allows different slopes
summary(model)

# The coefficient on "group" (or group: x interaction) gives the RDD estimate
```

Why the Other Options Are Incorrect

- **Second option** (`lm(y ~ group + x)`): Assumes **parallel slopes** on both sides → too restrictive, biased if slopes differ.

- **Third option** (`lm(y ~ x)`): Ignores the treatment group entirely → not an RDD model at all.

```

### Visual Interpretation (from the plot)

- Red points (control) left of cutoff ≈ 16.

- Cyan points (treatment) right of cutoff.

- Vertical lines show bandwidth window.

- We fit separate (or interacted) lines within the gray bandwidth area.

In regression discontinuity design, what conclusion can we make about individuals in different treatment groups with forcing variable values close to the cutpoint?

Individuals with forcing variable values close to the cutpoint are used in a sharp RDD analysis while individuals with forcing variable values far from the cutpoint are used in a fuzzy RDD analysis.

Individuals in different treatment groups with forcing variable values close to the cutpoint will be more similar to one another with respect to confounding variables than individuals further from the cutpoint.

👏 Correct!

Individuals with forcing variable values close to the cutpoint have a low probability of being in the treatment group whereas individuals with forcing variable values far from the cutpoint have a high probability of being in the treatment group.

Using treatment assignment as an instrument for the treatment received, we could use instrumental variable estimation to get a causal treatment effect in a situation where there is not perfect compliance with the treatment assignment.

The cells in the table below illustrate what the received treatment would be under different treatment assignment scenarios for each type of complier. Notice that in this table, the type of complier column has not been filled in.

| | Instrument Value | |
|---|---|---|
| Type of Complier | Assigned to Control | Assigned to Treatment |
| A | Treatment | Treatment |
| B | Treatment | Control |
| C | Control | Treatment |
| D | Control | Control |

Match the type of compliance to the appropriate row of the table using your knowledge of the different combinations of treatment assignment and treatment received. Then fill in the blanks to make the statement true.

A: ____

B: ____

C: ____

D: ____

In instrumental variable estimation, an individual's true compliance type ____ be determined. Only one combination of treatment assignment and treatment received is observed.

Never-taker | can | Always-taker | Defier | cannot | Complier

Click or drag and drop to fill in the blank

Using treatment assignment as an instrument for the treatment received, we could use instrumental variable estimation to get a causal treatment effect in a situation where there is not perfect compliance with the treatment assignment.

The cells in the table below illustrate what the received treatment would be under different treatment assignment scenarios for each type of complier. Notice that in this table, the type of complier column has not been filled in.

| | Instrument Value | |
|---|---|---|
| Type of Complier | Assigned to Control | Assigned to Treatment |
| A | Treatment | Treatment |
| B | Treatment | Control |
| C | Control | Treatment |
| D | Control | Control |

Match the type of compliance to the appropriate row of the table using your knowledge of the different combinations of treatment assignment and treatment received. Then fill in the blanks to make the statement true.

A: Always-taker

B: Defier

C: Complier

D: Never-taker

In instrumental variable estimation, an individual's true compliance type cannot be determined. Only one combination of treatment assignment and treatment received is observed.

You got it!

Suppose we are interested in using a regression discontinuity design analysis to determine the effect of a tutoring program on final exam performance. Students take an entrance exam and those who score 60% or lower get a free tutor for one year. Students then take the final exam at the end of the year. A student passes the final exam if they score at least a 75%. The following variables were measured for a group of students:

- `age`: student age (years);
- `entrance_score`: entrance exam score (0% - 100%);
- `final_score`: final exam score (0% - 100%);
- `tutor`: tutoring status (tutor vs. no tutor);

The bandwidth used to subset data in regression discontinuity design may be selected algorithmically. Fill in the code below to calculate the optimal IK bandwidth for this scenario.

```
# import library
library(rdd)
# determine bandwidth
IKbandwidth(
  X = entrance_score ,
  Y = final_score ,
  cutpoint = 60
)
```

You got it!

Imagine that we have access to a dataset named `df` which contains several variables:

- `outcome`: the outcome of interest;
- `treat`: the treatment variable;
- `instrument`: the instrument variable;

Fill in the code below to fit an appropriate two-stage least squares (2SLS) regression model for instrumental variable estimation.

```
# import library
library(AER)
# 2SLS model
ivreg(
  formula = outcome ~ treat | instrument ,
  data = df
)
```

You got it!

Regression discontinuity design uses only a subset of the full data. Which of the following statements about the causal treatment effect we can estimate using RDD is true?

- We can estimate the average treatment effect (ATE) for the full data.
- We can only estimate the treatment effect for the subset of the full data, known as the local average treatment effect (LATE).
- Results of a regression discontinuity design (RDD) analysis are easily generalizable because RDD estimates the local average treatment effect (LATE).
- We cannot estimate any causal treatment effect.

Regression discontinuity design uses only a subset of the full data. Which of the following statements about the causal treatment effect we can estimate using RDD is true?

- We can estimate the average treatment effect (ATE) for the full data.
- We can only estimate the treatment effect for the subset of the full data, known as the local average treatment effect (LATE).

Correct!

- Results of a regression discontinuity design (RDD) analysis are easily generalizable because RDD estimates the local average treatment effect (LATE).
- We cannot estimate any causal treatment effect.